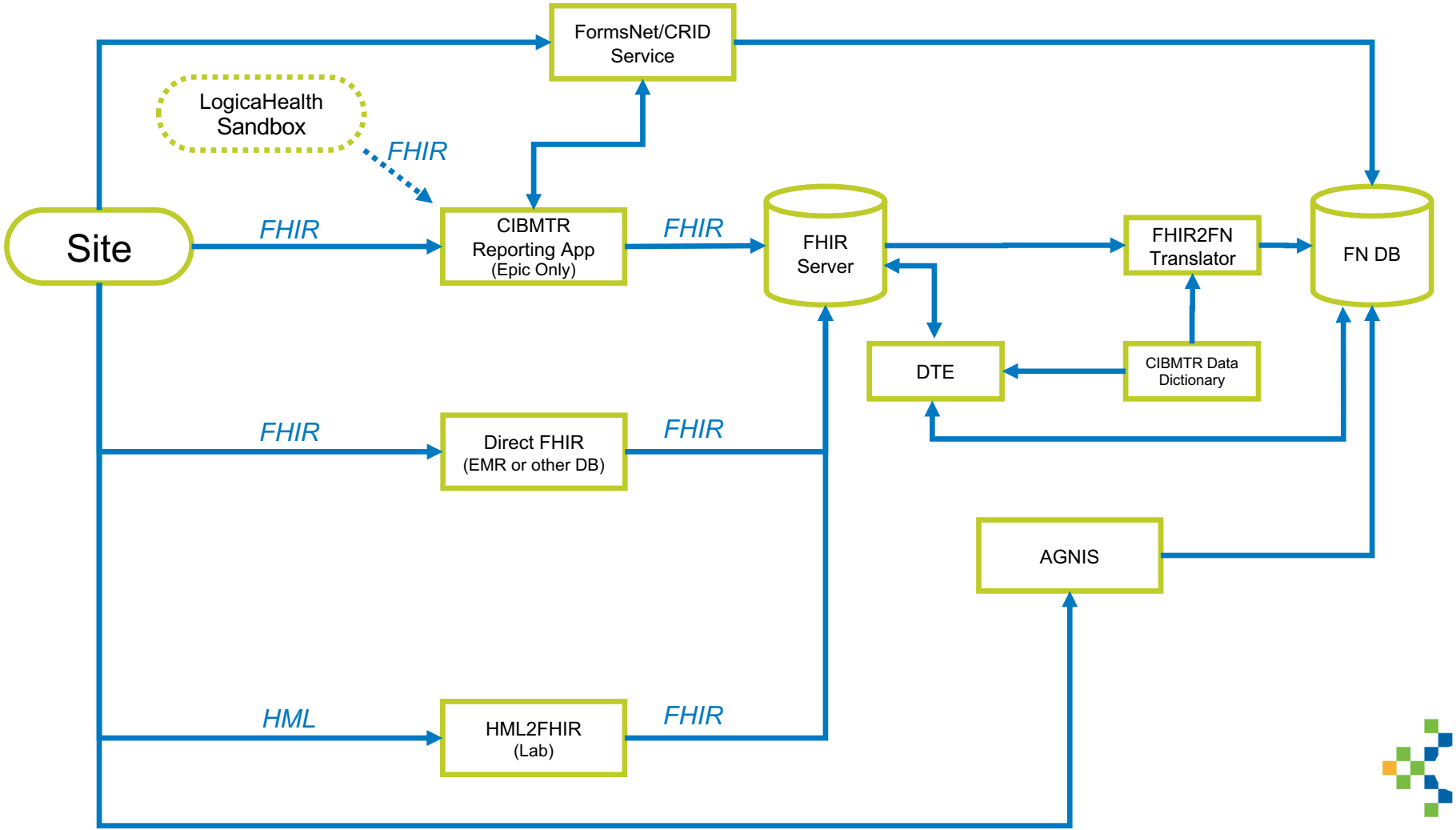
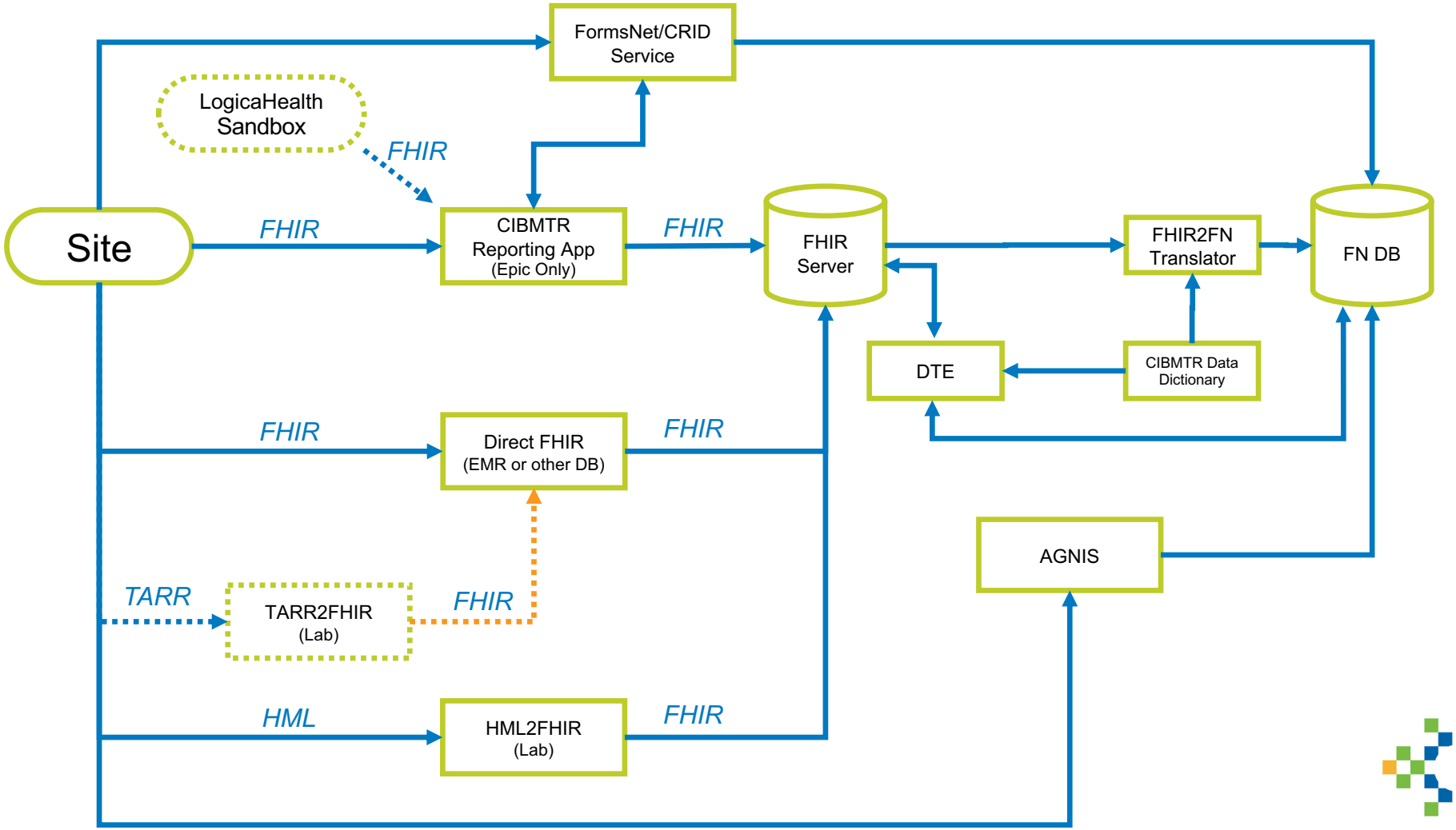


HLA to FHIR: TARR2FHIR

8/6/21

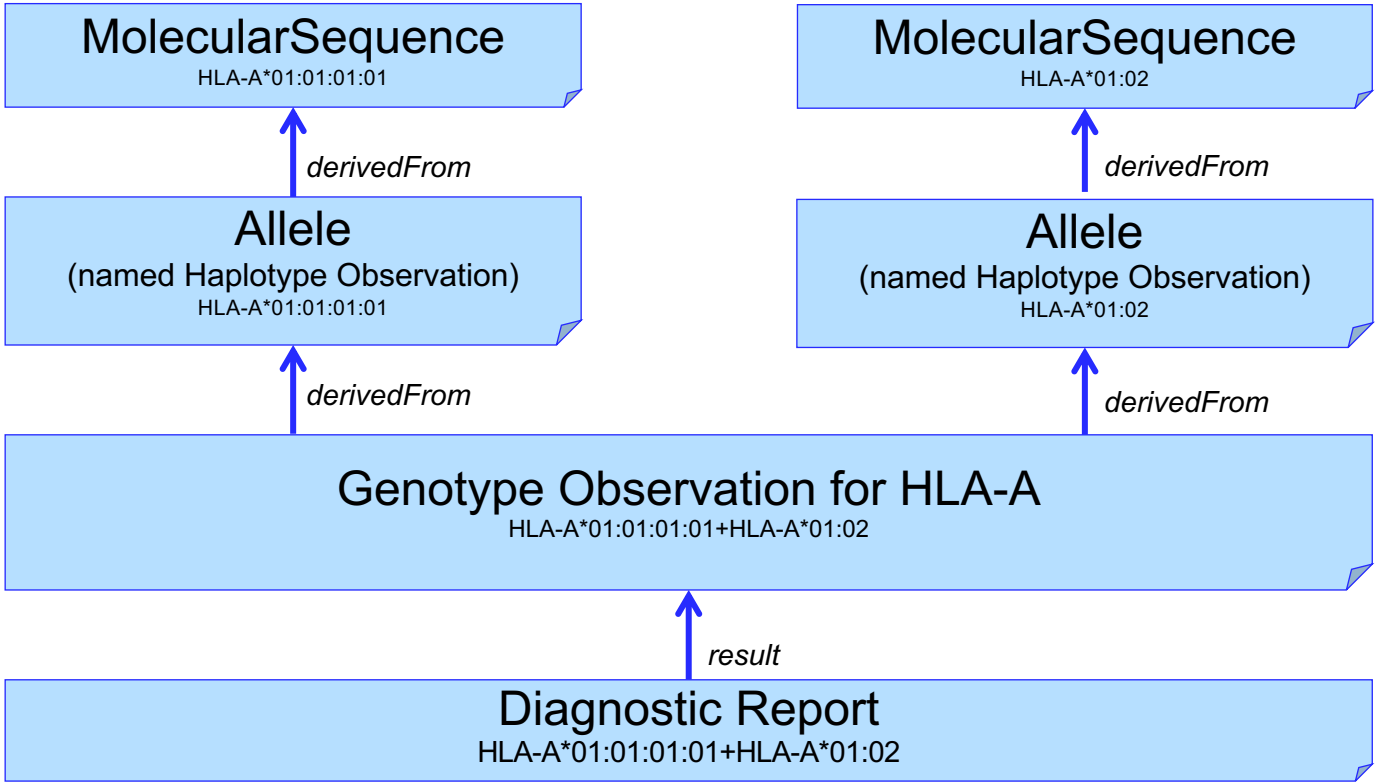

FormsNet/CRID Service
LogicaHealth Sandbox
FHIR
Site
FHIR
CIBMTR Reporting App (Epic Only)
FHIR
FHIR Server
FHIR2FN Translator
FN DB
DTE
CIBMTR Data Dictionary
FHIR
Direct FHIR (EMR or other DB)
FHIR
AGNIS
HML
HML2FHIR (Lab)
FHIR


FormsNet/CRID Service
LogicaHealth Sandbox
FHIR
Site
FHIR
CIBMTR Reporting App (Epic Only)
FHIR
FHIR Server
FHIR2FN Translator
FN DB
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CIBMTR Data Dictionary
FHIR
Direct FHIR (EMR or other DB)
FHIR
AGNIS
TARR
TARR2FHIR (Lab)
FHIR
HML
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FHIR

ONC Sync for Genes: NMDP/CIBMTR

- Phase 1
  - Development of HLA Genotyping report using FHIR
  - Conversion of HML to FHIR
  - FHIR STU3
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  - FHIR R4, prior to publication of HL7 Genomics Reporting IG STU1
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  - Development of HLA Reporting FHIR IG
    - Derived from & informed by Genomics Reporting IG STU1
  - Direct FHIR submission by HLA typing lab
    - Versiti - https://www.versiti.org/

Bundle contents for one HLA gene



HLA Reporting IG

- 1st draft found at http://fhir.nmdp.org/ig/hla-reporting
- Derived from HL7 Genomics Reporting IG
- FHIR Shorthand (FSH) & Sushi to develop



What's in the IG?

- Profiles
  - HLA Summary Report
  - HLA Genotype Observation
  - HLA Allele Observation
  - HLA Molecular Sequence
- Value Sets / Code Systems
  - HLA Gene Name Value Set
  - HLA HGNC GeneID CodeSystem
  - Genotype List String Code CodeSystem
- Extension
  - HLA Genotype Summary
- Examples!



HLA-REPORTING\Home - FHIR

fhir.nmdp.org/ig/hla-reporting/

HLA Reporting Implementation Guide

0.0.1 - Draft

IG Home | Table of Contents | Background | Artifact Index | Support

Table of Contents > Home

This page is part of the HLA Reporting Implementation Guide (v0.0.1: Trial Use: 1) based on FHIR R4. This is the current published version. For a full list of available versions, see the Directory of published versions

1 Home

Human leukocyte antigen (HLA) genotyping is fundamental for research and clinical practice in immunogenetics and histocompatibility. This HLA Reporting Implementation Guide (IG) describes how to report HLA genotyping results using HL7 FHIR. It relies heavily on the Genomics Reporting IG developed by the HL7 Clinical Genomics Workgroup, constraining profiles for GeneticsReport, Genotype and Haplotype. It also constrains the MolecularSequence resource to allow reporting sequencing results together with novel variants.

Contents:

- Profiles in this IG
- Building a report
- HLA - Reporting Ambiguity
- HLA - LOINC and additional codings
- HLA - Examples

1.1 Profiles in this IG

- HLA Summary Report
- HLA Genotype
- HLA Allele
- HLA Molecular Sequence

1.2 Building a report

A complete HLA genotyping report is typically contained in a FHIR transaction Bundle containing individual resources conforming to the profiles described in this IG. Examples of full transaction bundles included below. Minimally, it would contain an HLA Summary Report and one or more HLA Genotype observations.

1.2.1 Summary Report and Genotype Observation

In a simple case, the HLA Summary Report refers to an HLA Genotype observation for each HLA gene. The HLA alleles found in the genotype are assigned to alleles found in the the Immuno Polymorphism Database (IPD) through the international ImMunoGeneTics (IMGT) project (found here) and expressed as a Genotype List String (GLString) using a GLString Code.

If it is necessary to report the molecular sequence that was used to derive the HLA Genotype observation, one or more HLA Sequence profiles can be used. This is shown in Fig 1.

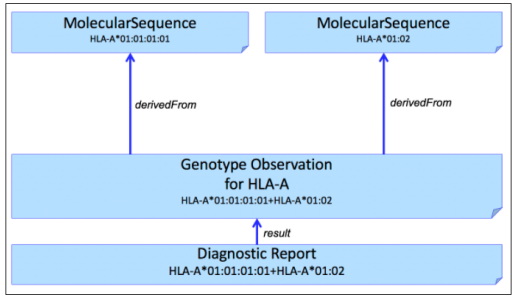
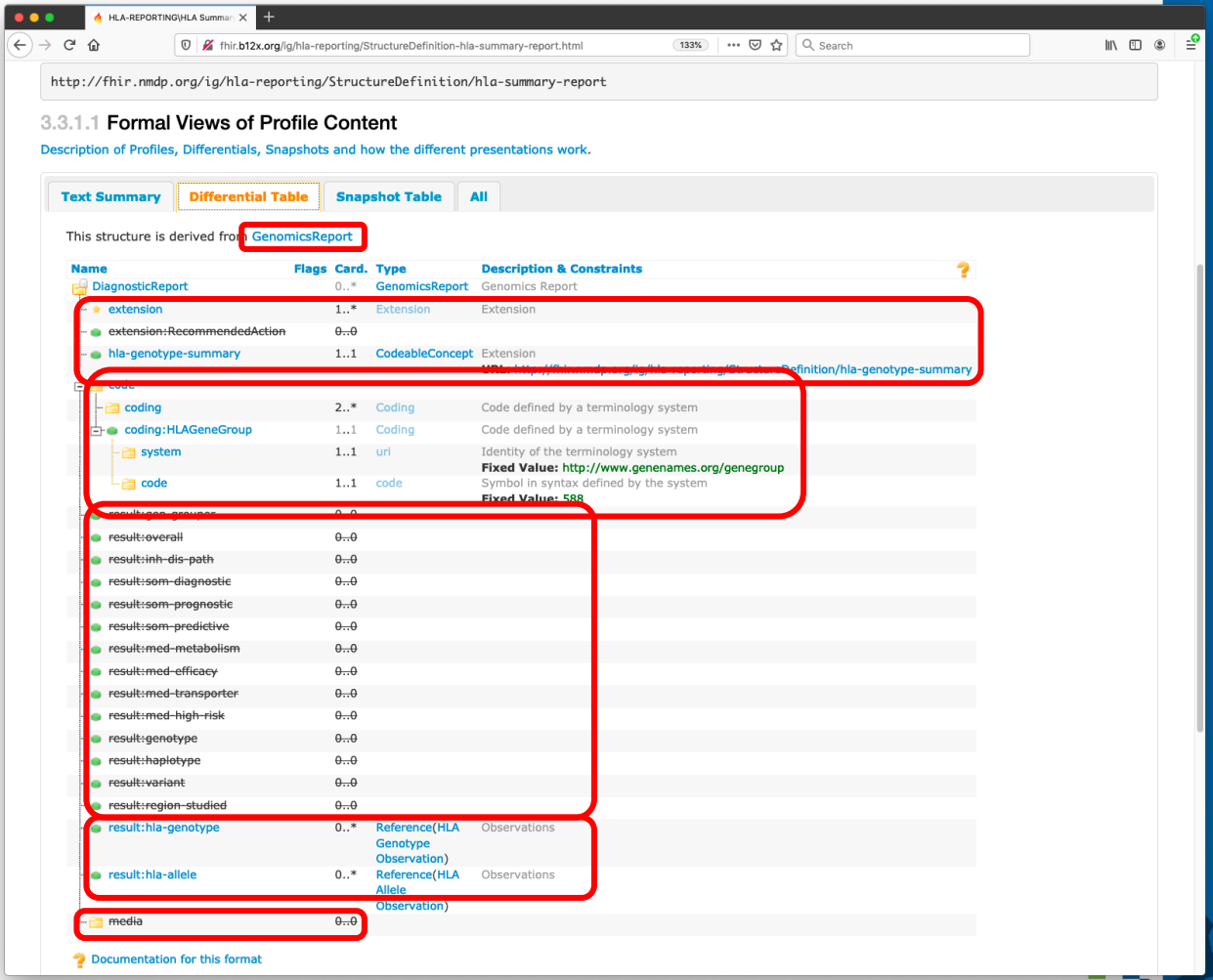


Figure 1

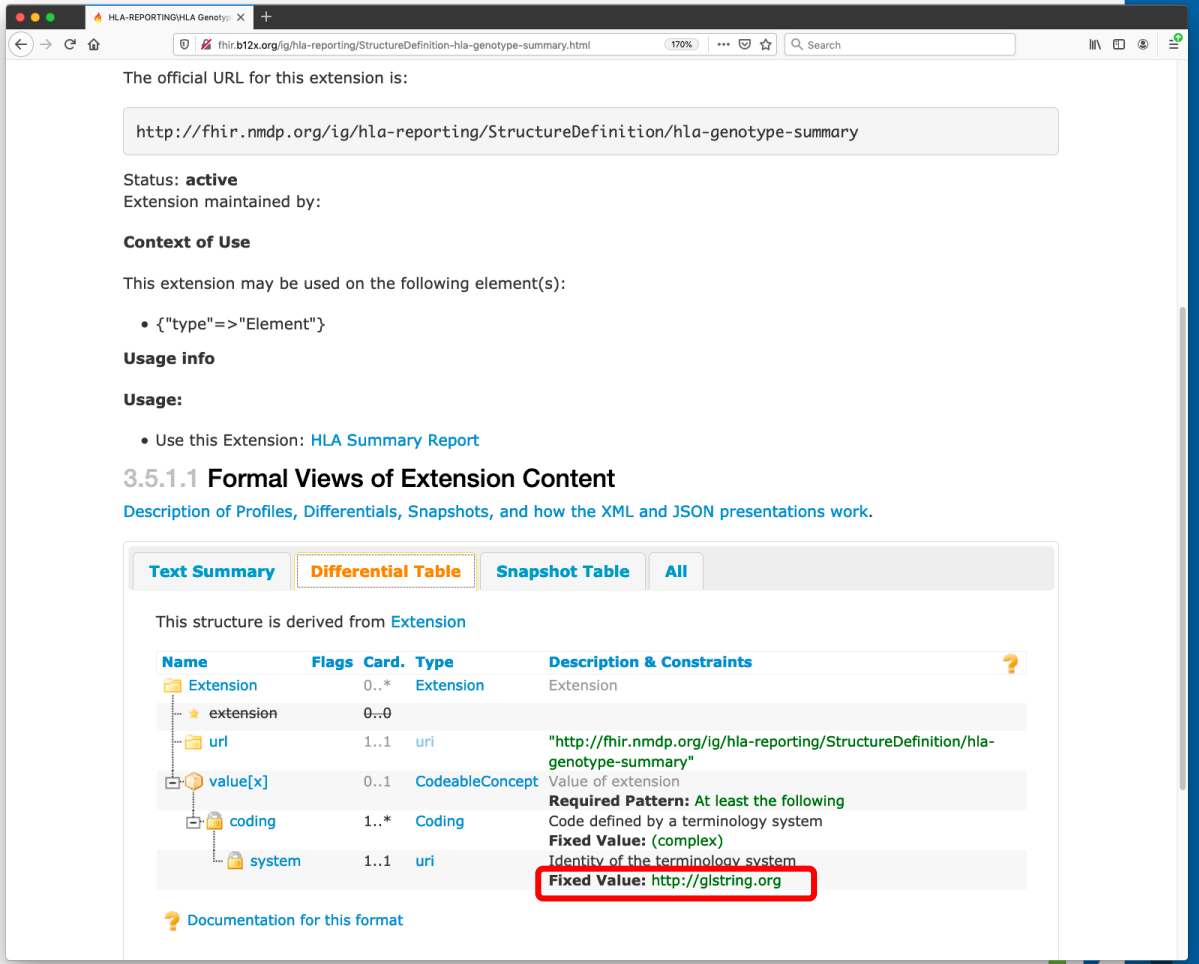
HLA Summary Report

- derived from GenomicsReport
- constrained
  - code
  - results
  - media
- added extension to report multi-locus genotype



HLA Genotype Summary Extension

- in HLA Summary Report
- multi-locus genotypes



1.6.2.4 HLA - Reporting Ambiguity

HLA allele assignment is often ambiguous. Genotype ambiguity results from an inability to establish chromosomal phase between identified polymorphisms. Allele ambiguity results when the polymorphisms that distinguish alleles fall outside of the regions assessed by the genotyping system.

Allele ambiguity is often captured using NMDP Multiple Allele Codes. Because of limitations of this system, including introducing further ambiguity into the report, the GL String is preferred. This format uses a hierarchical set of operators to describe the relationships between alleles, lists of possible alleles, phased alleles, genotypes, lists of possible genotypes, and multilocus unphased genotypes, without losing typing information or increasing typing ambiguity. To use this format, the GL String Code system can be used, which embeds this GL String into a format containing the gene system, and version of the nomenclature used within the GL String.

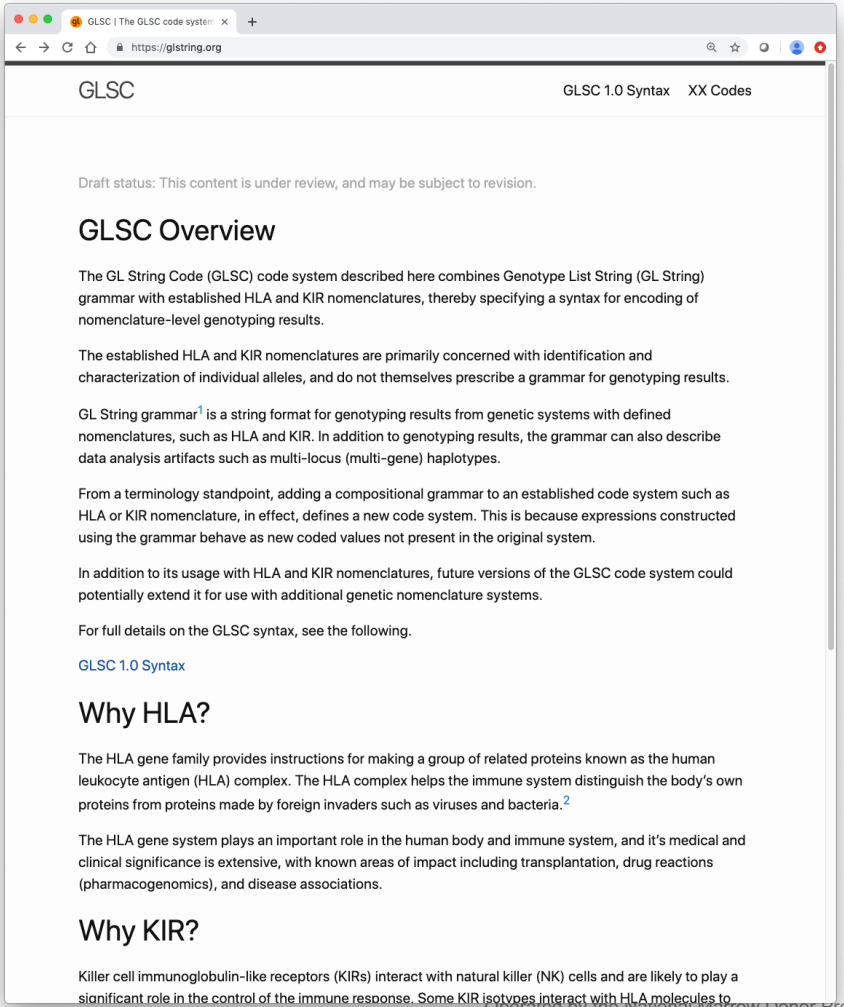
For example, this valueCodeableConcept would be used to describe a genotype for HLA-B, based on the 3.31.0 release of the IMGT-IPD/HLA database, that is ambiguous for one of the alleles in the genotype (HLA-B*07:02:01:01/HLA-B*07:02:01:03), and unambiguous for the other (HLA-B*13:02:01:01).

```
<valueCodeableConcept>
      <coding>
          <system value="https://glstring.org"/>
          <version value="1.0"/>
          <code value="hla#3.31.0#HLA-B*07:02:01:01/HLA-B*07:02:01:03+HLA-B*13:02:01:01"/>
      </coding>
</valueCodeableConcept>
```

http://glstring.org

The GL String Code (GLSC) code system described here combines Genotype List String (GL String) grammar with established HLA and KIR nomenclatures, thereby specifying a syntax for encoding of nomenclature-level genotyping results.

In addition to its usage with HLA and KIR nomenclatures, future versions of the GLSC code system could potentially extend it for use with additional genetic nomenclature systems.

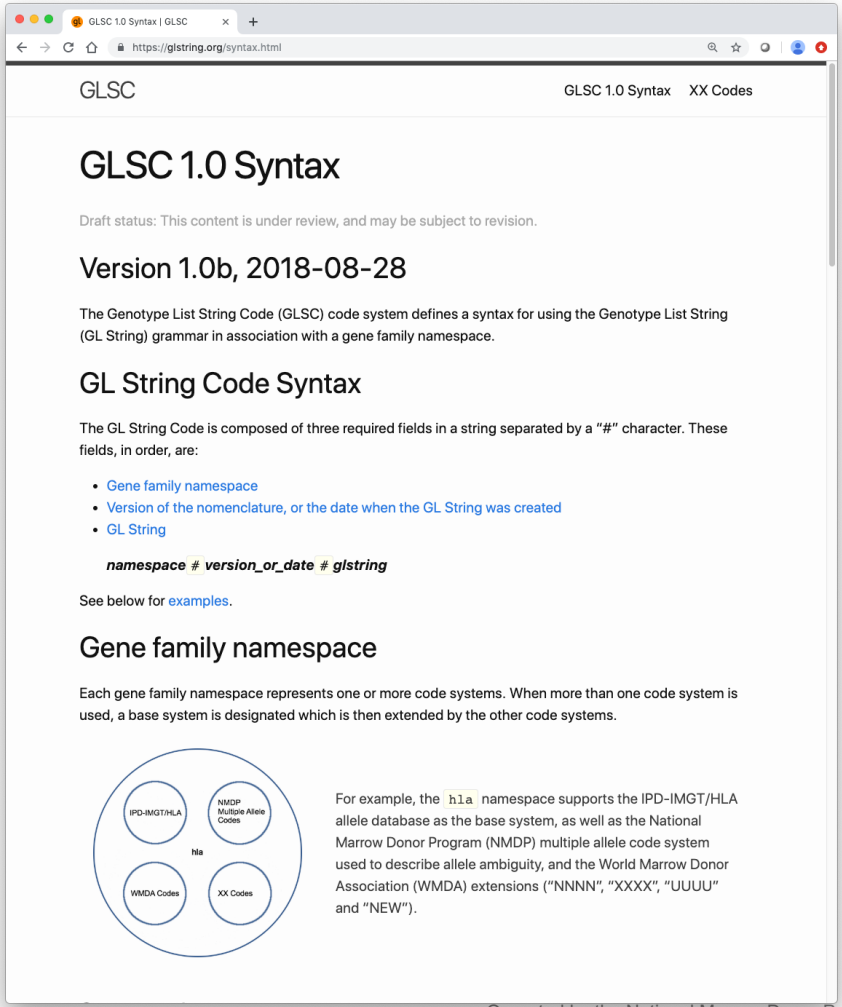


http://glstring.org

The GL String Code is composed of three required fields in a string separated by a “#” character. These fields, in order, are:

- Gene family namespace
- Version of the nomenclature, or the date when the GL String was created
- GL String

namespace#version_or_date#glstring



GL String Code Examples

Base nomenclature only

- `hla#3.25.0#HLA-A*01:01:01:01/HLA-A*01:02+HLA-A*24:02:01:01`
- `hla#3.29.0#HLA-DRB1*03:01:02~HLA-DRB5*01:01:01`
- `hla#3.33.0#HLA-A*02:69+HLA-A*23:30|HLA-A*02:302+HLA-A*23:26/HLA-A*23:39`
- `hla#2018-06#HLA-A*02:69+HLA-A*23:30`
- `kir#2.3.0#KIR3DL2*001+KIR3DL2*007|KIR3DL2*006+KIR3DL2*010`

NMDP Multiple Allele Code (MAC) Designations

- `hla#2018-03-01#HLA-DPB1*04:ANKZX+HLA-DPB1*04:FNVS`

MAC and WMDA extensions

- `hla#3.25.0#HLA-DPB1*04:ANKZX+HLA-DPB1*04:FNVS^HLA-DRB3*XXXX`

as a valueCodeableConcept.coding within a HL7 FHIR Observation...

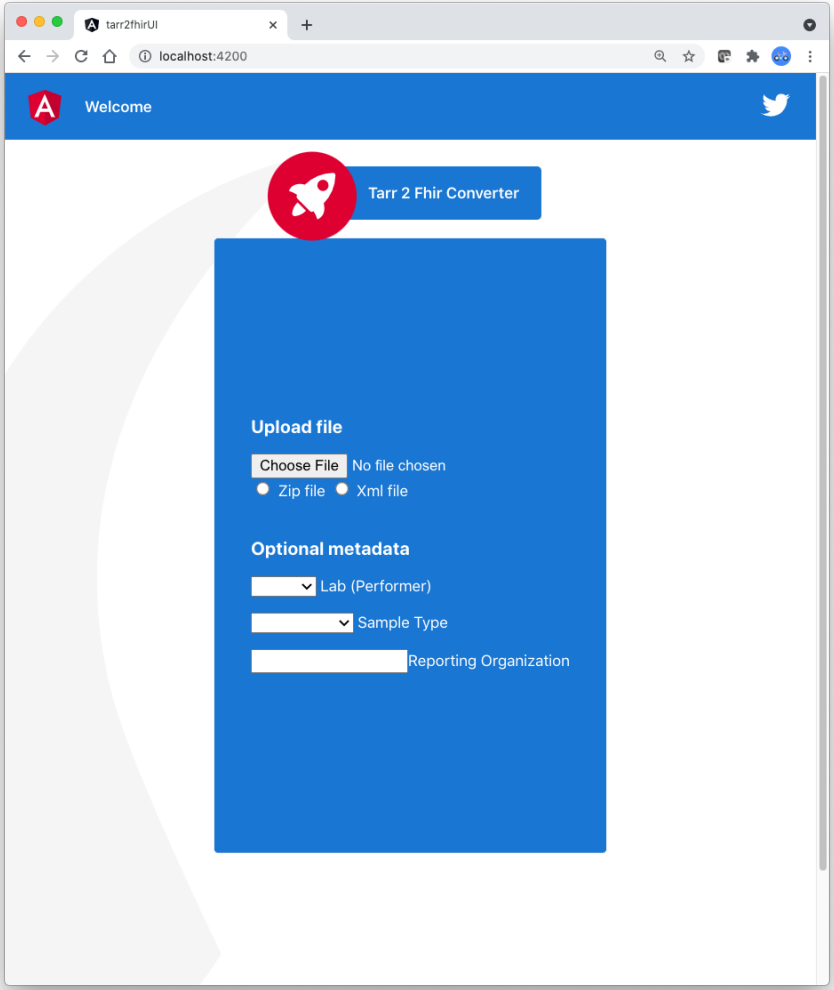
```
<valueCodeableConcept>
  <coding>
    <system value="http://glstring.org"/>
    <version value="1.0"/>
    <code value="hla#3.25.0#HLA-A*01:01:01:01/HLA-A*01:02+HLA-A*24:02:01:01"/>
  </coding>
</valueCodeableConcept>
```

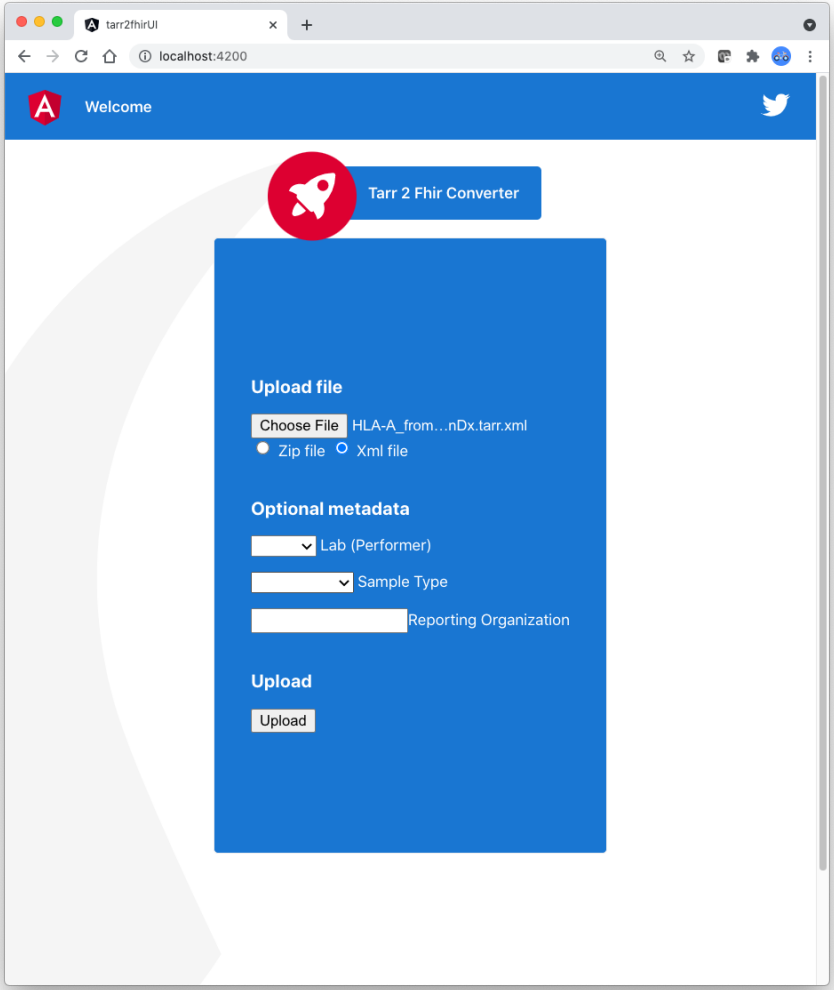
ONC Sync for Genes: NMDP/CIBMTR

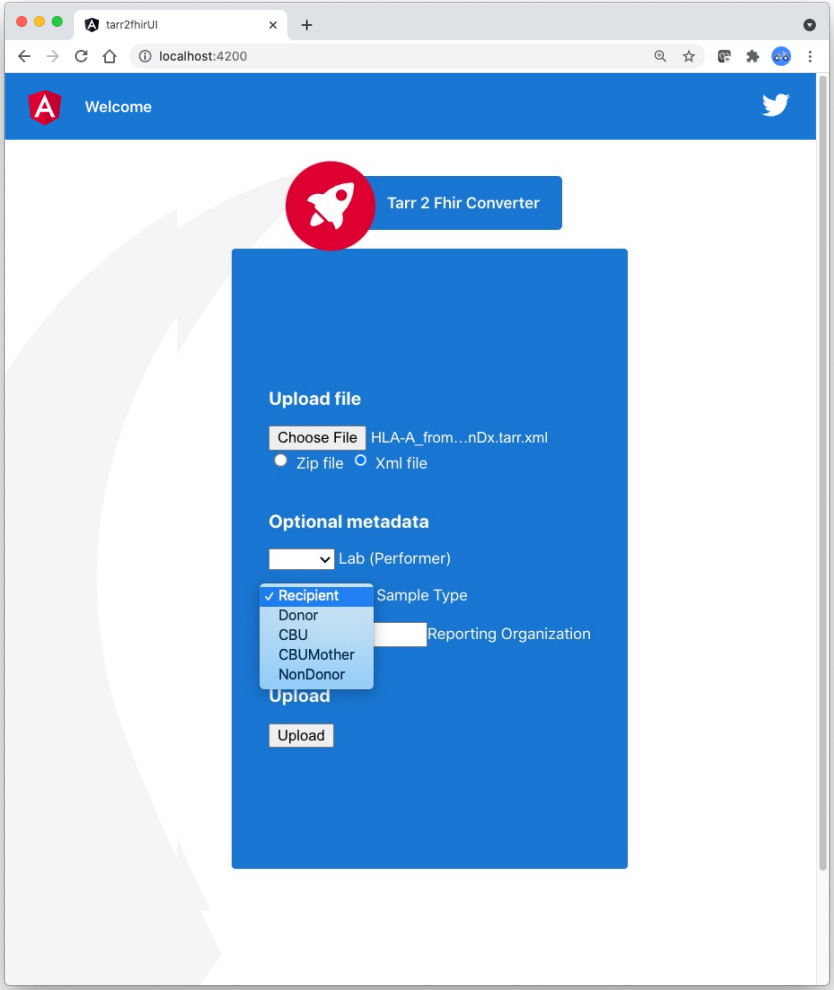
- ✓ Phase 1
  - ✓ Development of HLA Genotyping report using FHIR
  - ✓ Conversion of HML to FHIR
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  - ✓ pilot with Transplant Center partner (Moffit)
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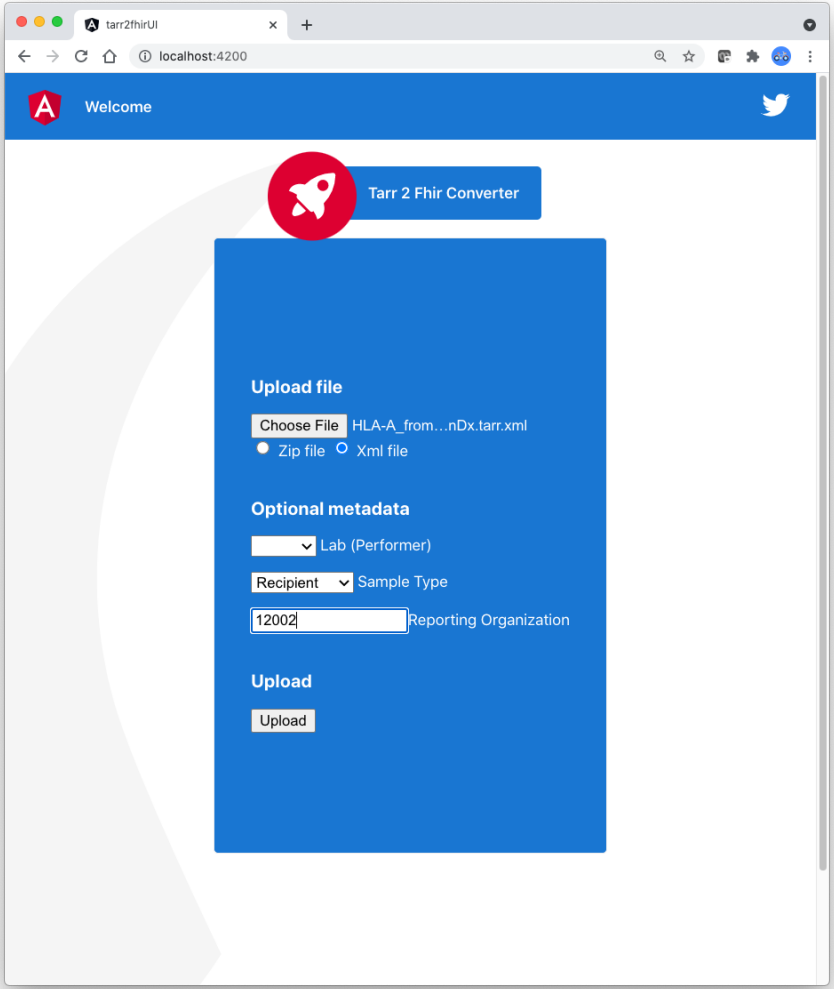
TARR2FHIR

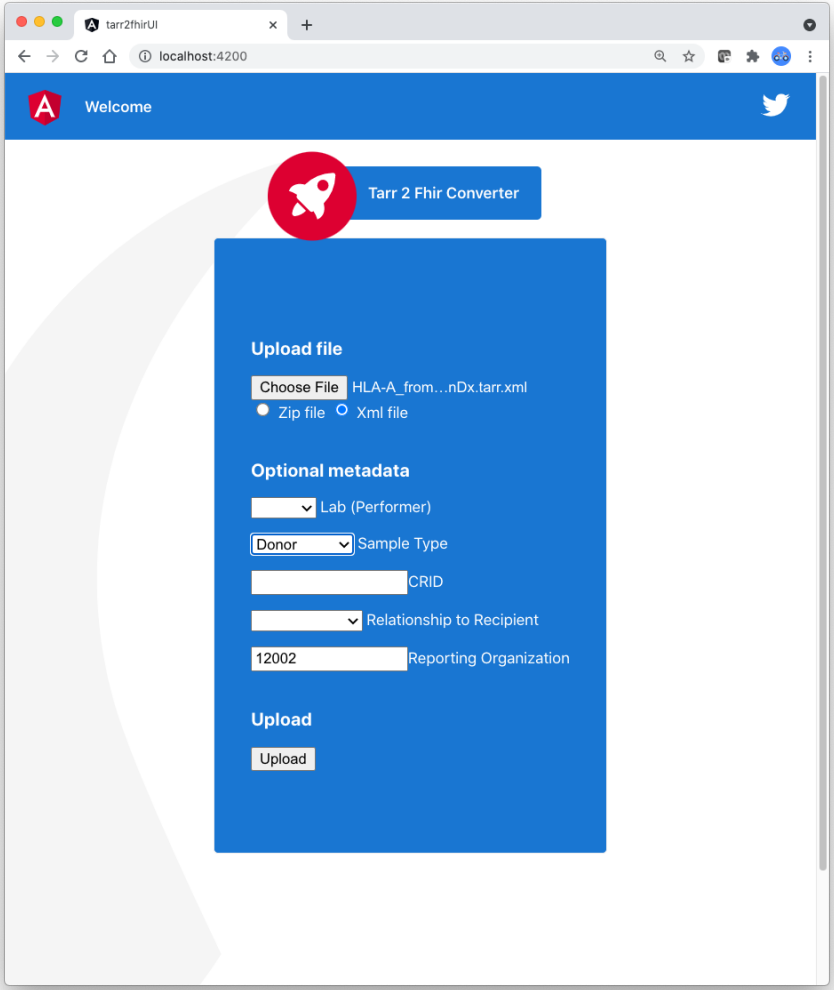
- https://github.com/nmdp-bioinformatics/tarr2fhir
- Converts GenDx NGSEngine® XML (TARR) to FHIR transaction Bundle that conforms to the HLA Reporting Implementation Guide
- Input – single TARR XML file for a single locus, or a zip file containing multiple TARR XML files for a single subject.
- Output - FHIR transaction Bundle containing
  - A single HLA Summary (profile based on GenomicsReport) for a single subject
  - HLA Genotype (profile based on Genotype Observation), one per HLA gene
  - HLA Allele (profile based on Haplotype Observation), one per allele
  - HLA MolecularSequence (profile based on MolecularSequence), one per sequence reported
  - A single Provenance resource (to describe how the resources were created based on what input file and what software)
  - A single Device resource (to describe TARR2FHIR conversion software & version)

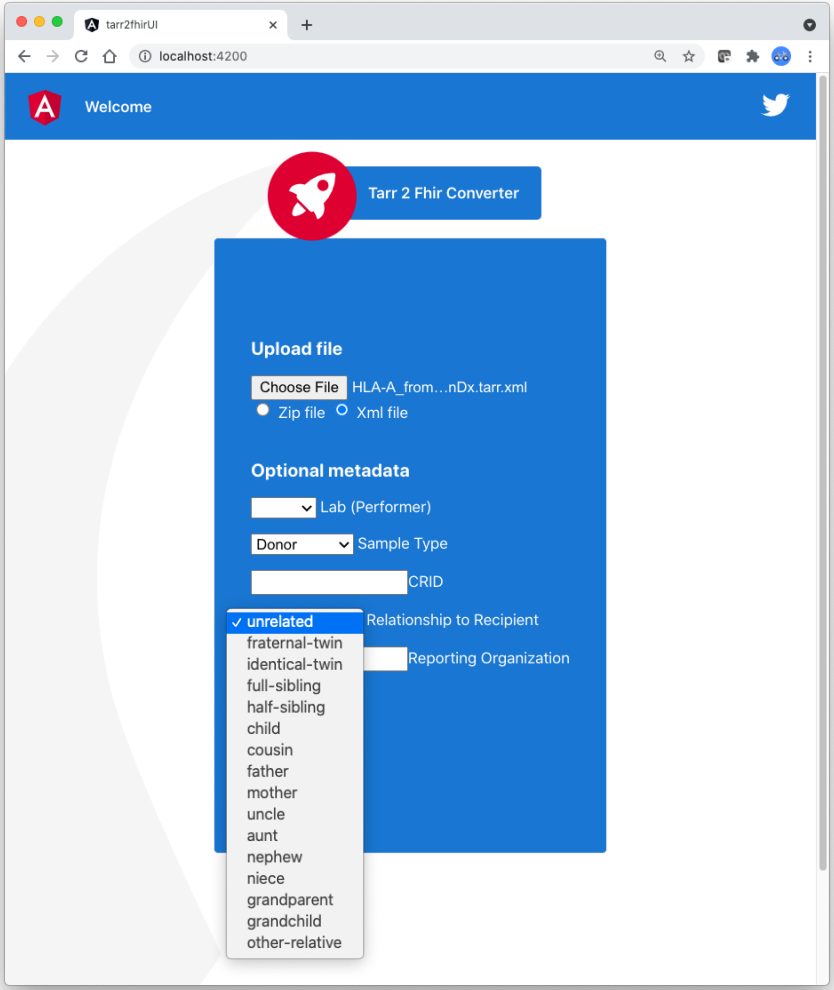

tarr2fhirUI
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Welcome
Tarr 2 Fhir Converter
Upload file
Choose File
No file chosen
Zip file
Xml file
Optional metadata
Lab (Performer)
Sample Type
Reporting Organization

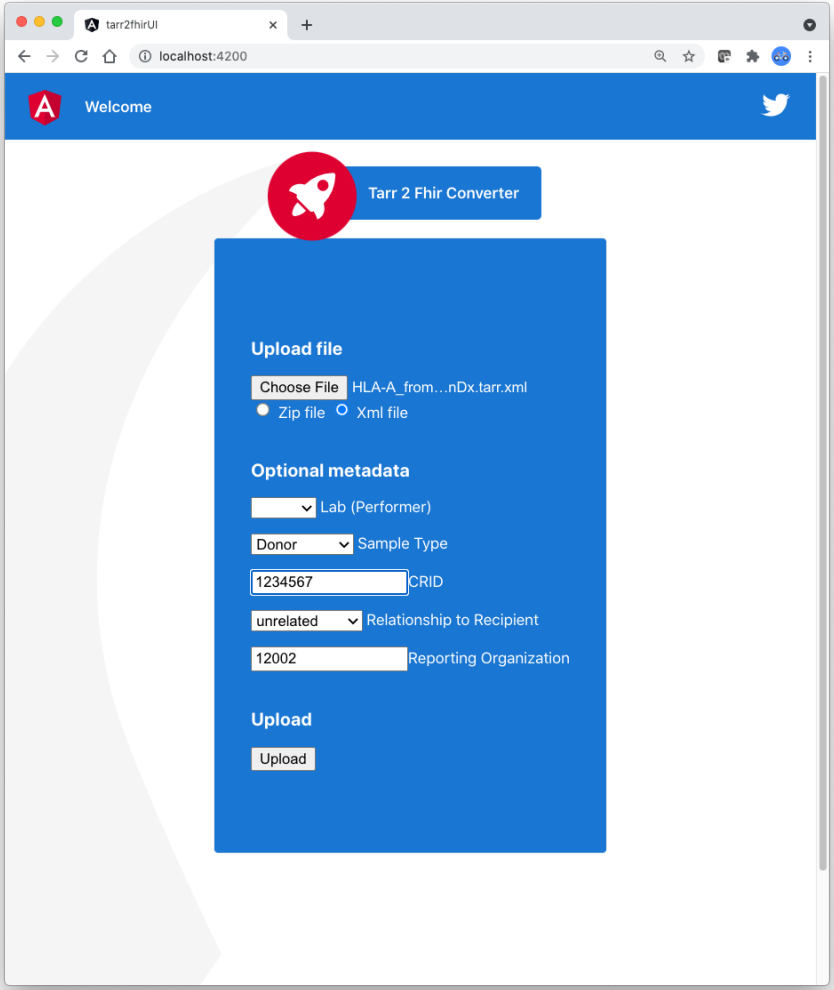

tarr2fhirUI
localhost:4200
Welcome
Tarr 2 Fhir Converter
Upload file
Choose File
HLA-A_from…nDx.tarr.xml
Zip file
Xml file
Optional metadata
Lab (Performer)
Sample Type
Reporting Organization
Upload
Upload

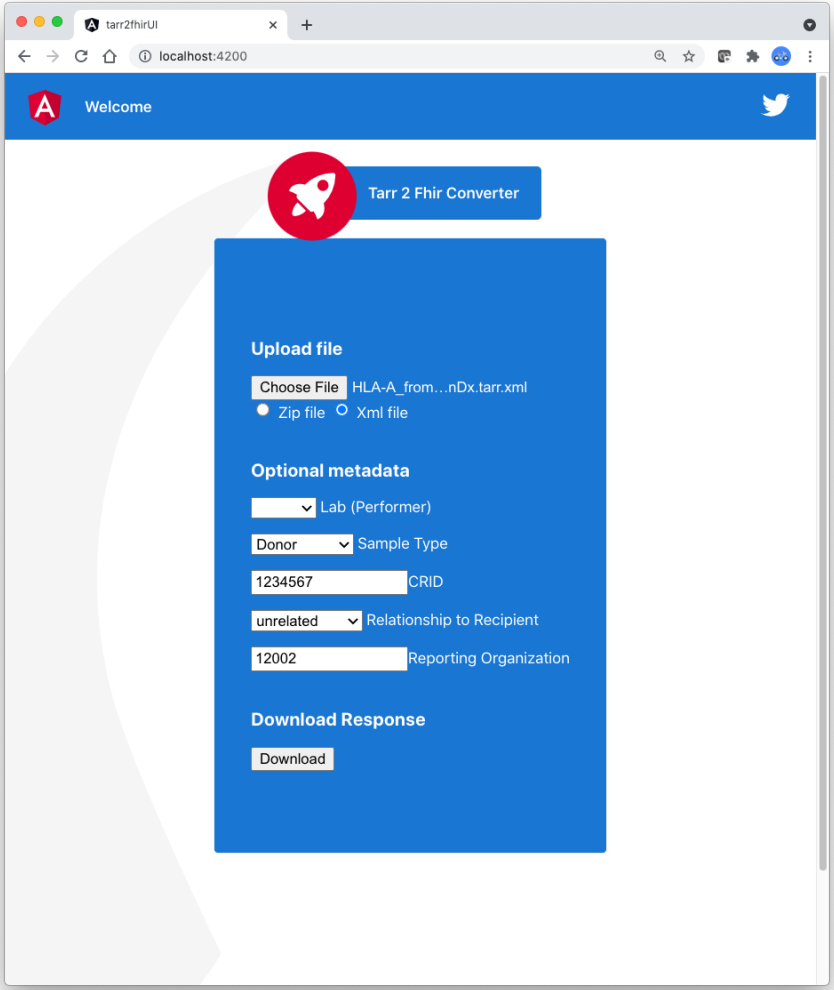

tarr2fhirUI
localhost:4200
Welcome
Tarr 2 Fhir Converter
Upload file
Choose File
HLA-A_from…nDx.tarr.xml
Zip file
Xml file
Optional metadata
Lab (Performer)
Recipient
Donor
CBU
CBUMother
NonDonor
Sample Type
Reporting Organization
Upload
Upload

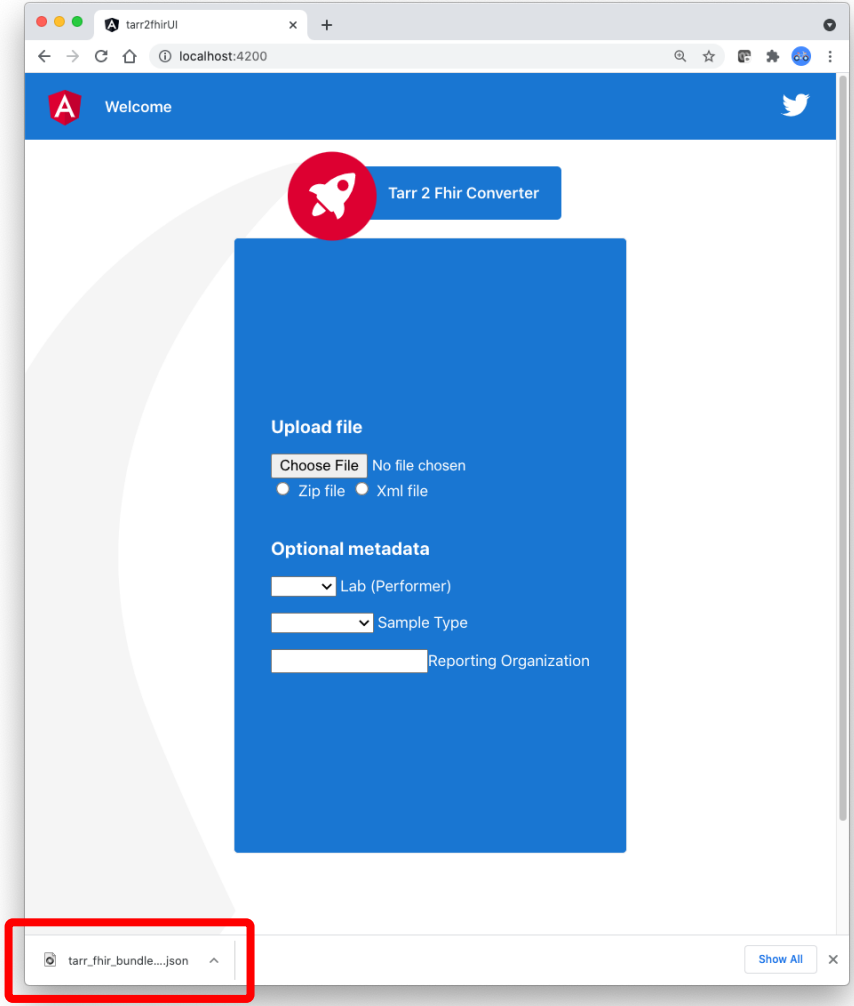

tarr2fhirUI
localhost:4200
Welcome
Tarr 2 Fhir Converter
Upload file
Choose File
HLA-A_from…nDx.tarr.xml
Zip file
Xml file
Optional metadata
Lab (Performer)
Recipient
Sample Type
12002
Reporting Organization
Upload
Upload


tarr2fhirUI
localhost:4200
Welcome
Tarr 2 Fhir Converter
Upload file
Choose File
HLA-A_from…nDx.tarr.xml
Zip file
Xml file
Optional metadata
Lab (Performer)
Donor
Sample Type
CRID
Relationship to Recipient
12002
Reporting Organization
Upload
Upload


tarr2fhirUI
localhost:4200
Welcome
Tarr 2 Fhir Converter
Upload file
Choose File
HLA-A_from...nDx.tarr.xml
Zip file
Xml file
Optional metadata
Lab (Performer)
Donor
Sample Type
CRID
unrelated
fraternal-twin
identical-twin
full-sibling
half-sibling
child
cousin
father
mother
uncle
aunt
nephew
niece
grandparent
grandchild
other-relative
Relationship to Recipient
Reporting Organization


tarr2fhirUI
localhost:4200
Welcome
Tarr 2 Fhir Converter
Upload file
Choose File
HLA-A_from…nDx.tarr.xml
Zip file
Xml file
Optional metadata
Lab (Performer)
Donor
Sample Type
1234567
CRID
unrelated
Relationship to Recipient
12002
Reporting Organization
Upload
Upload


tarr2fhirUI
localhost:4200
Welcome
Tarr 2 Fhir Converter
Upload file
Choose File
HLA-A_from…nDx.tarr.xml
Zip file
Xml file
Optional metadata
Lab (Performer)
Donor
Sample Type
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CRID
unrelated
Relationship to Recipient
12002
Reporting Organization
Download Response
Download

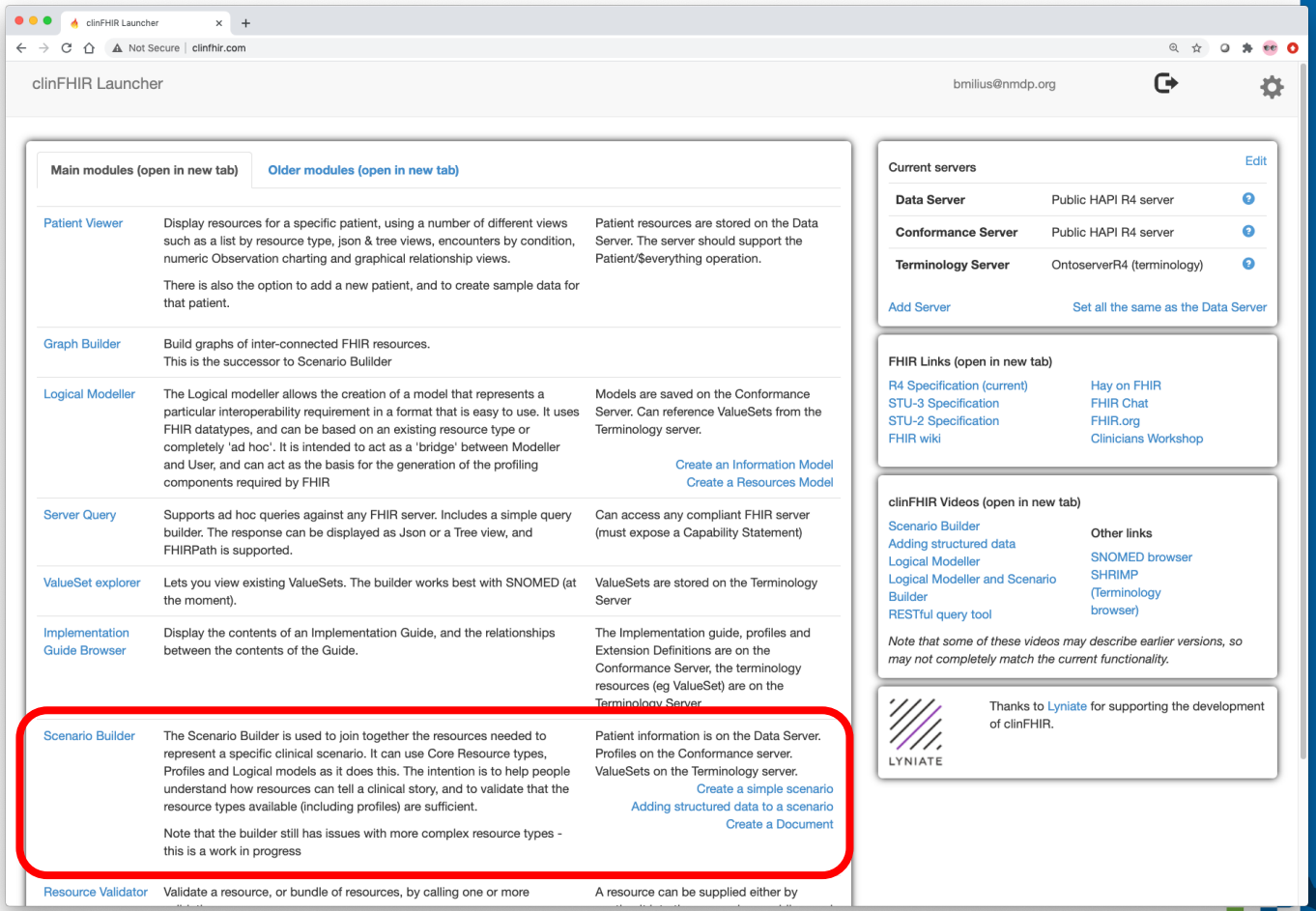

tarr2fhirUI
localhost:4200
Welcome
Tarr 2 Fhir Converter
Upload file
Choose File
No file chosen
Zip file
Xml file
Optional metadata
Lab (Performer)
Sample Type
Reporting Organization
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Show All

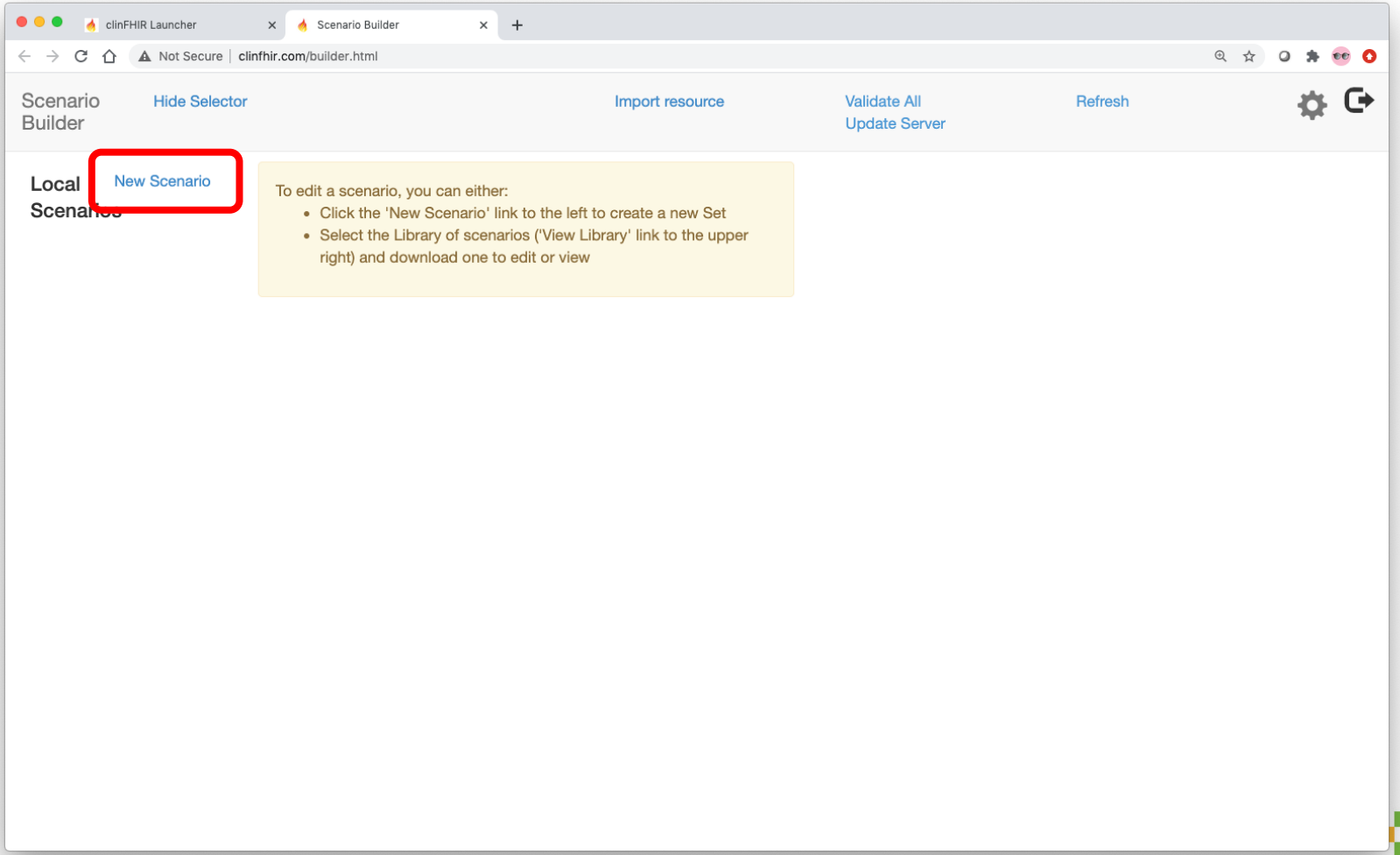

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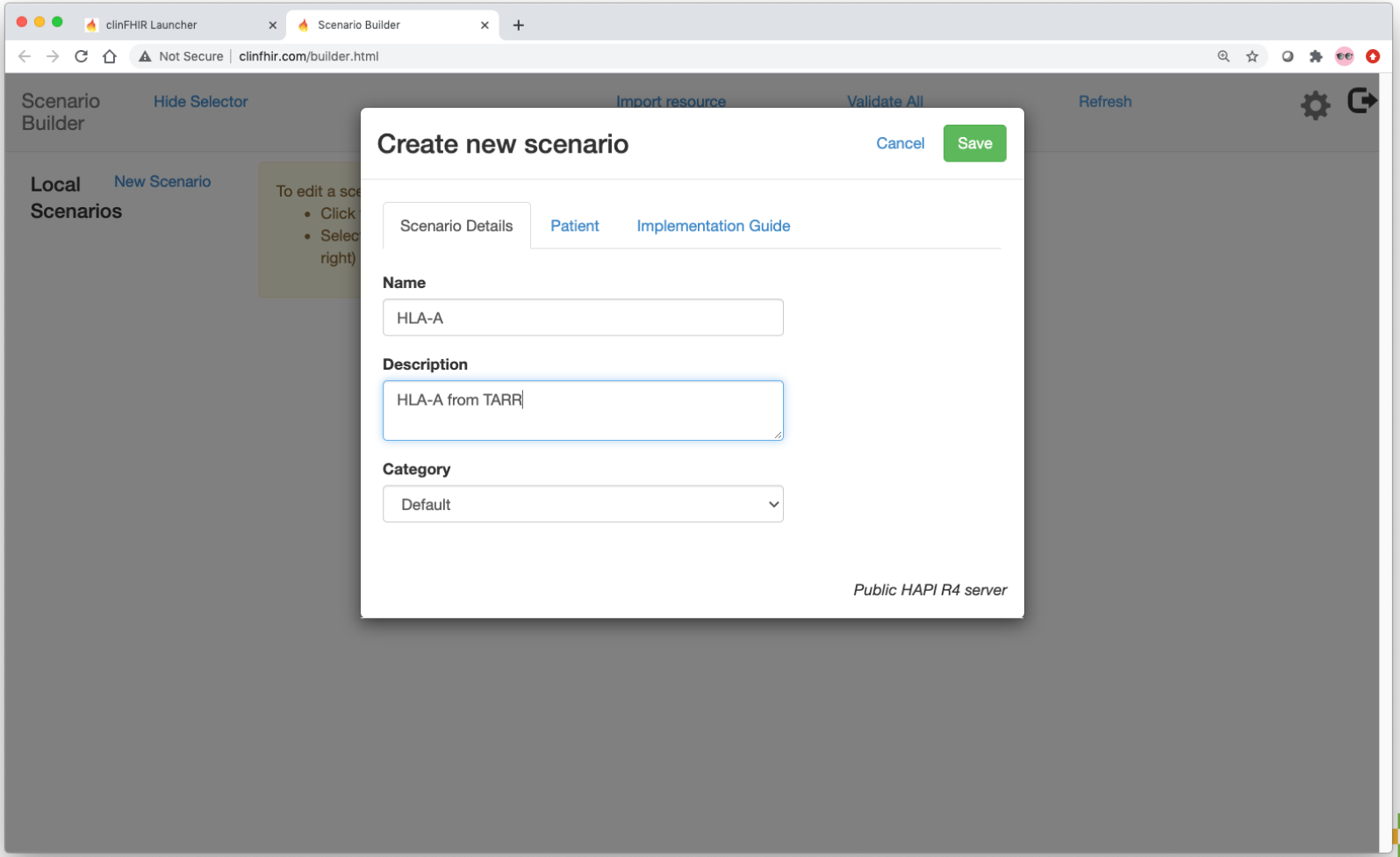
26

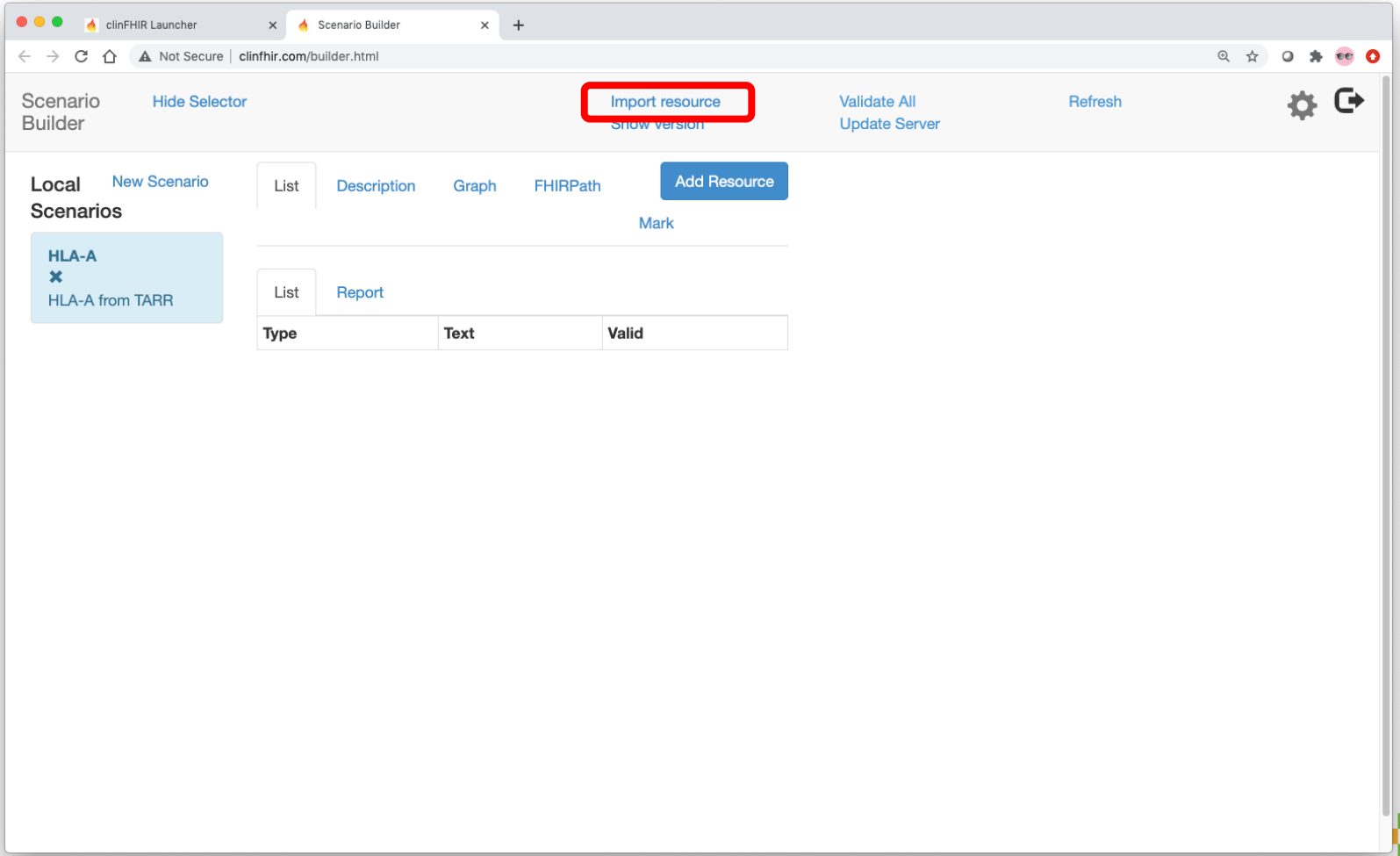
clinFHIR.com

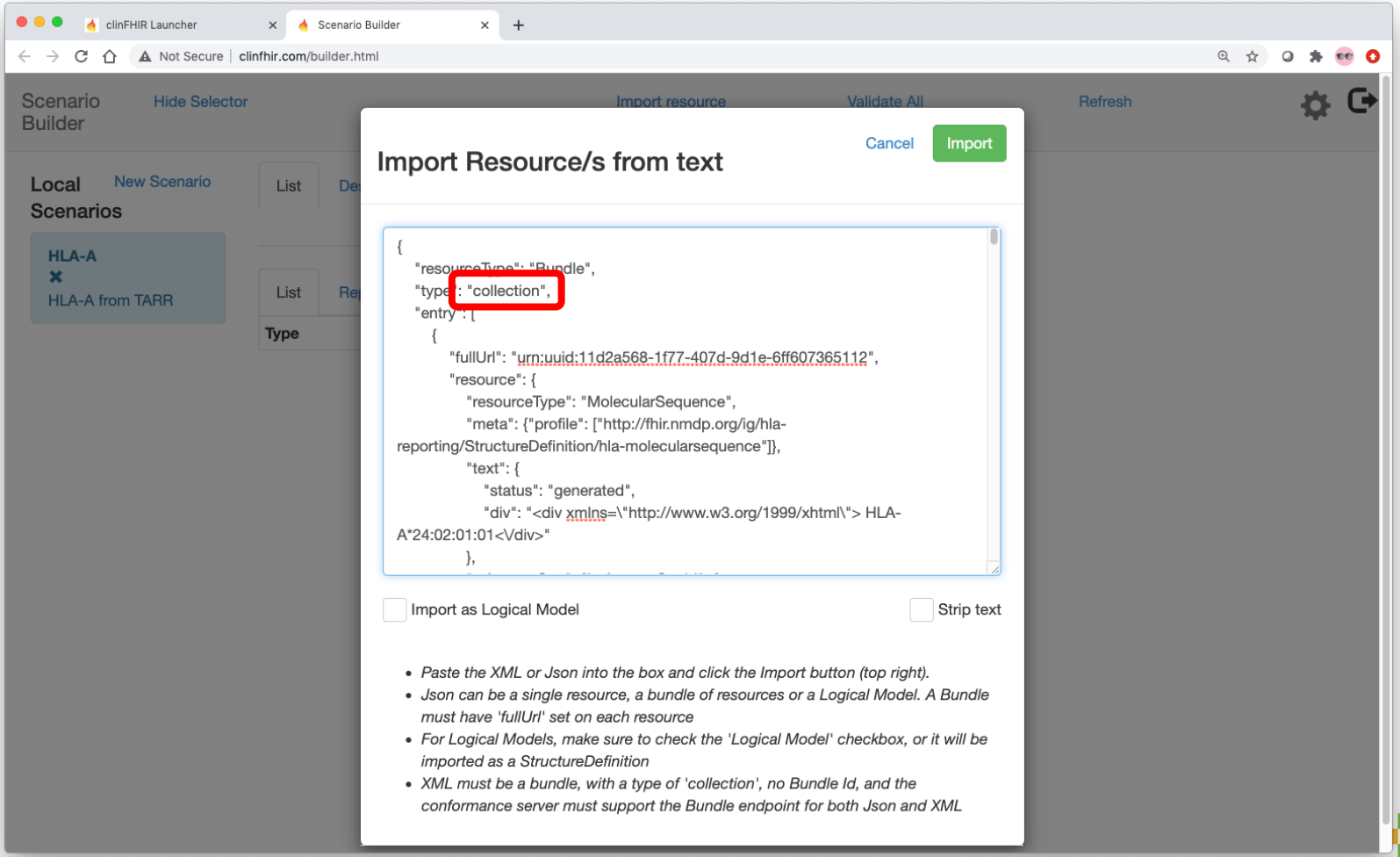
- view FHIR bundles using Scenario Builder

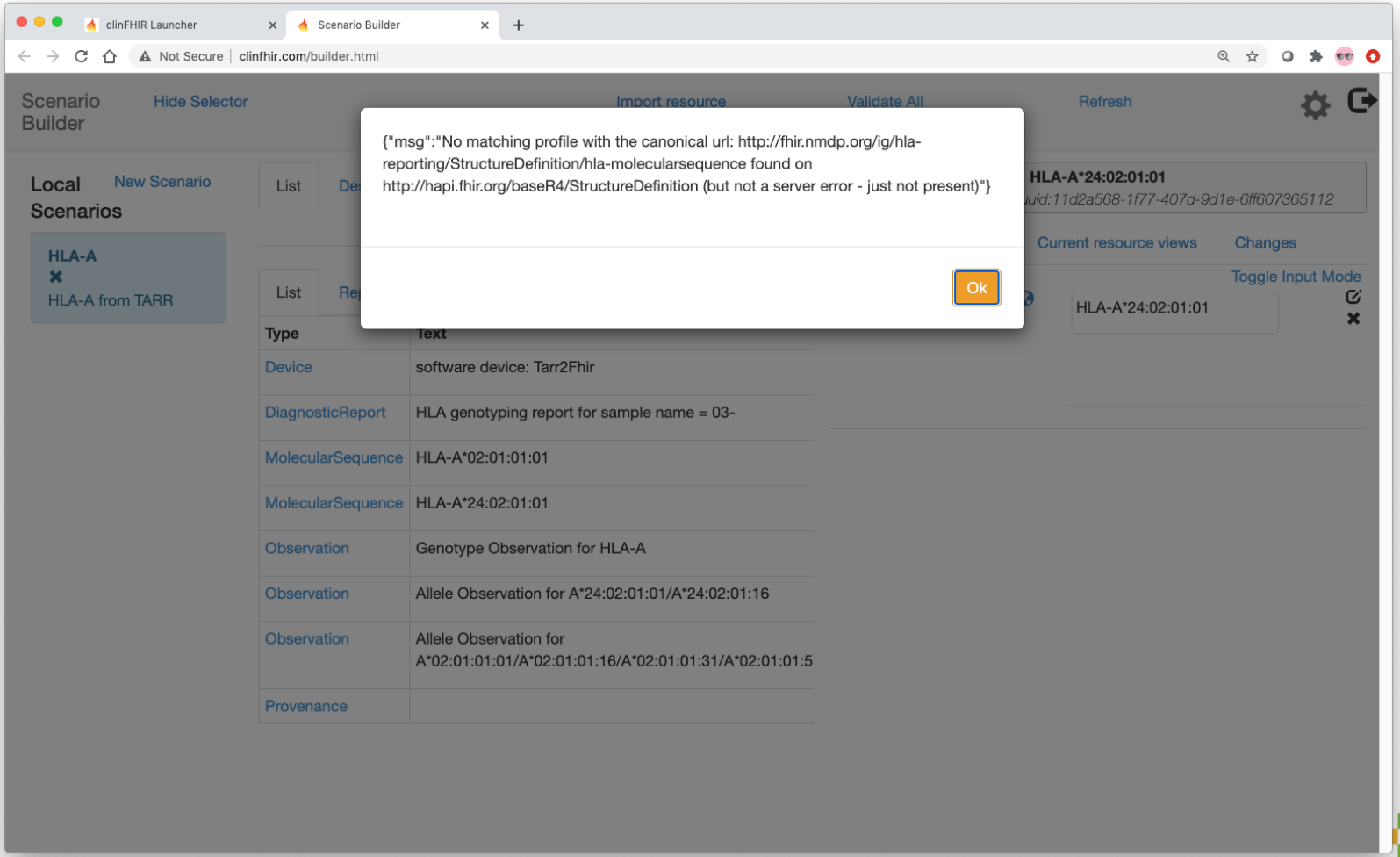


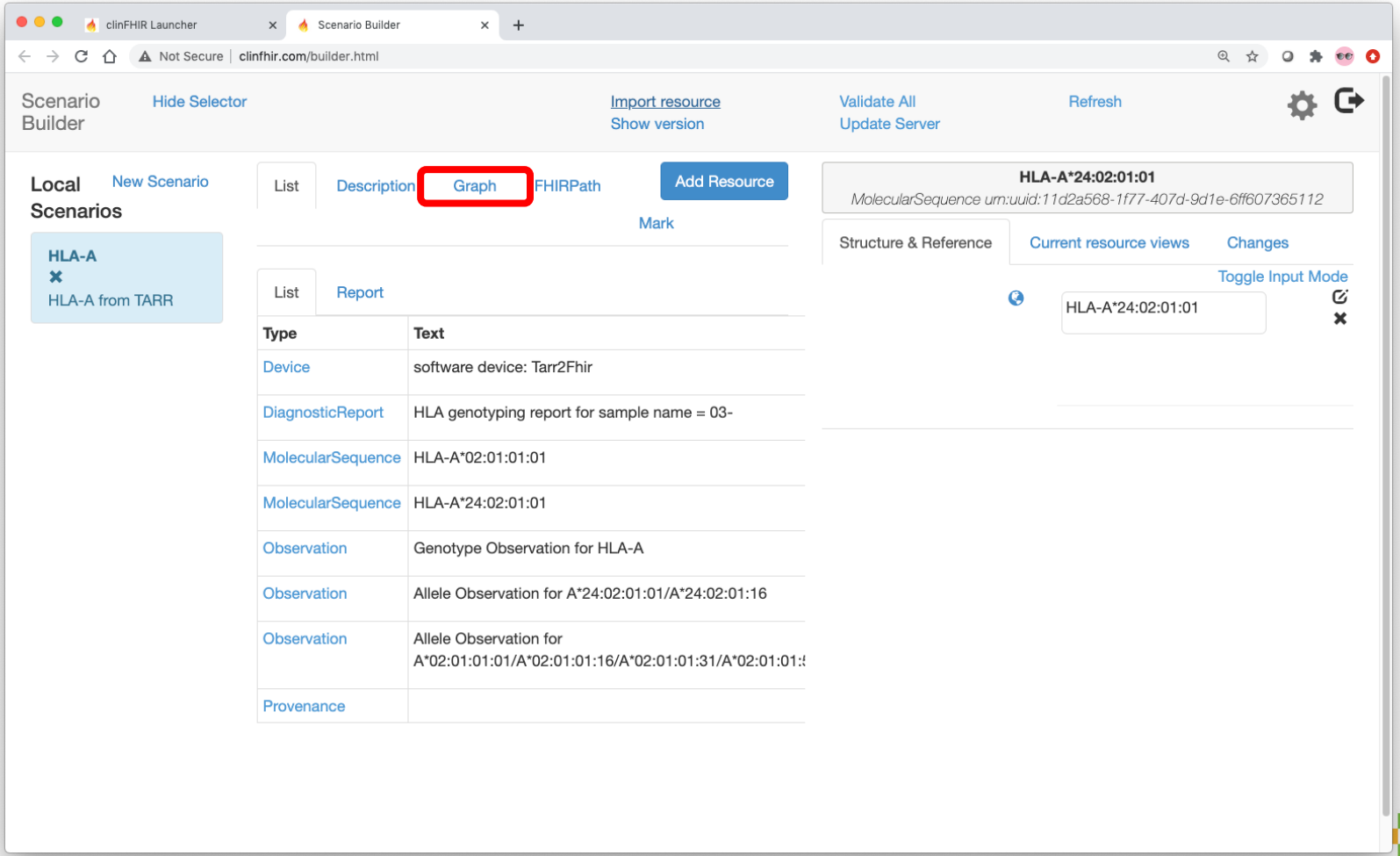

clinFHIR Launcher
Scenario Builder
Not Secure | clinfhir.com/builder.html
Scenario Builder
Hide Selector
Import resource
Validate All
Update Server
Refresh
Local Scenarios
New Scenario
To edit a scenario, you can either:
Click the 'New Scenario' link to the left to create a new Set
Select the Library of scenarios ('View Library' link to the upper right) and download one to edit or view

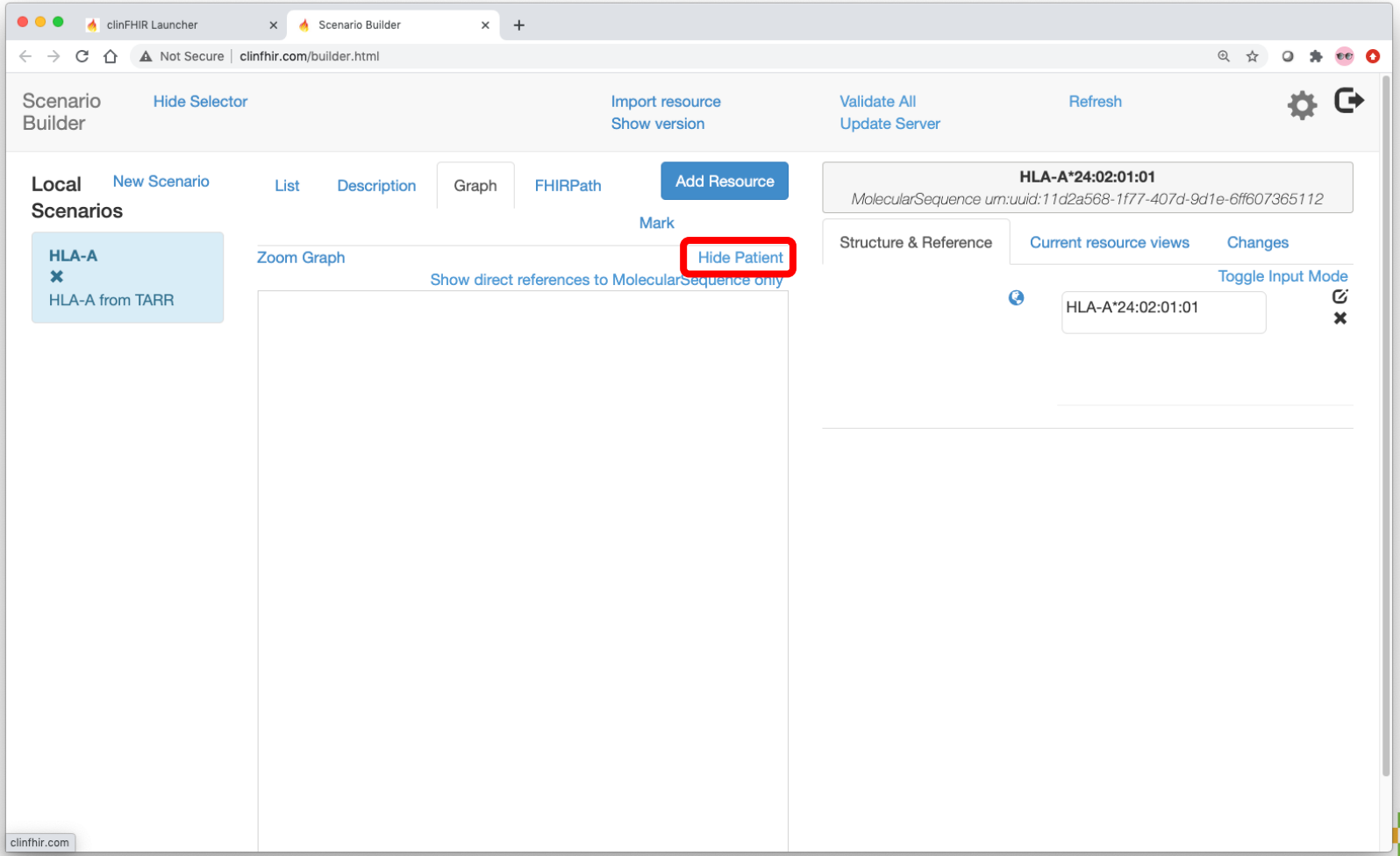

clinFHIR Launcher
Scenario Builder
Not Secure | clinfhir.com/builder.html
Scenario Builder
Hide Selector
Import resource
Validate All
Refresh
Local Scenarios
New Scenario
Create new scenario
Cancel
Save
Scenario Details
Patient
Implementation Guide
Name
HLA-A
Description
HLA-A from TARR
Category
Default
Public HAPI R4 server

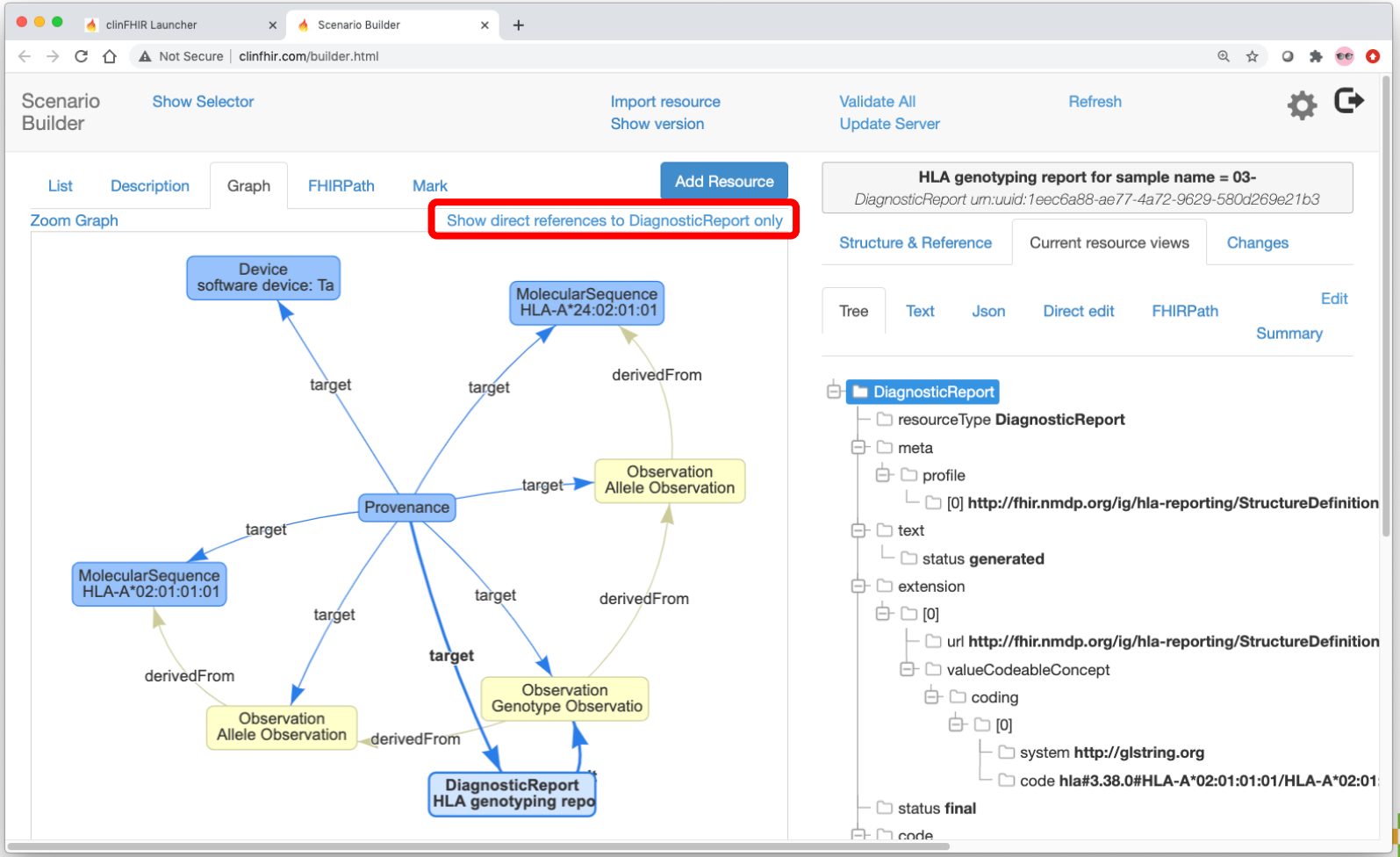

clinFHIR Launcher
Scenario Builder
Not Secure | clinfhir.com/builder.html
Scenario Builder
Hide Selector
Import resource
Show Version
Validate All
Update Server
Refresh
Local Scenarios
New Scenario
HLA-A
HLA-A from TARR
List
Description
Graph
FHIRPath
Add Resource
Mark
List
Report
Type
Text
Valid

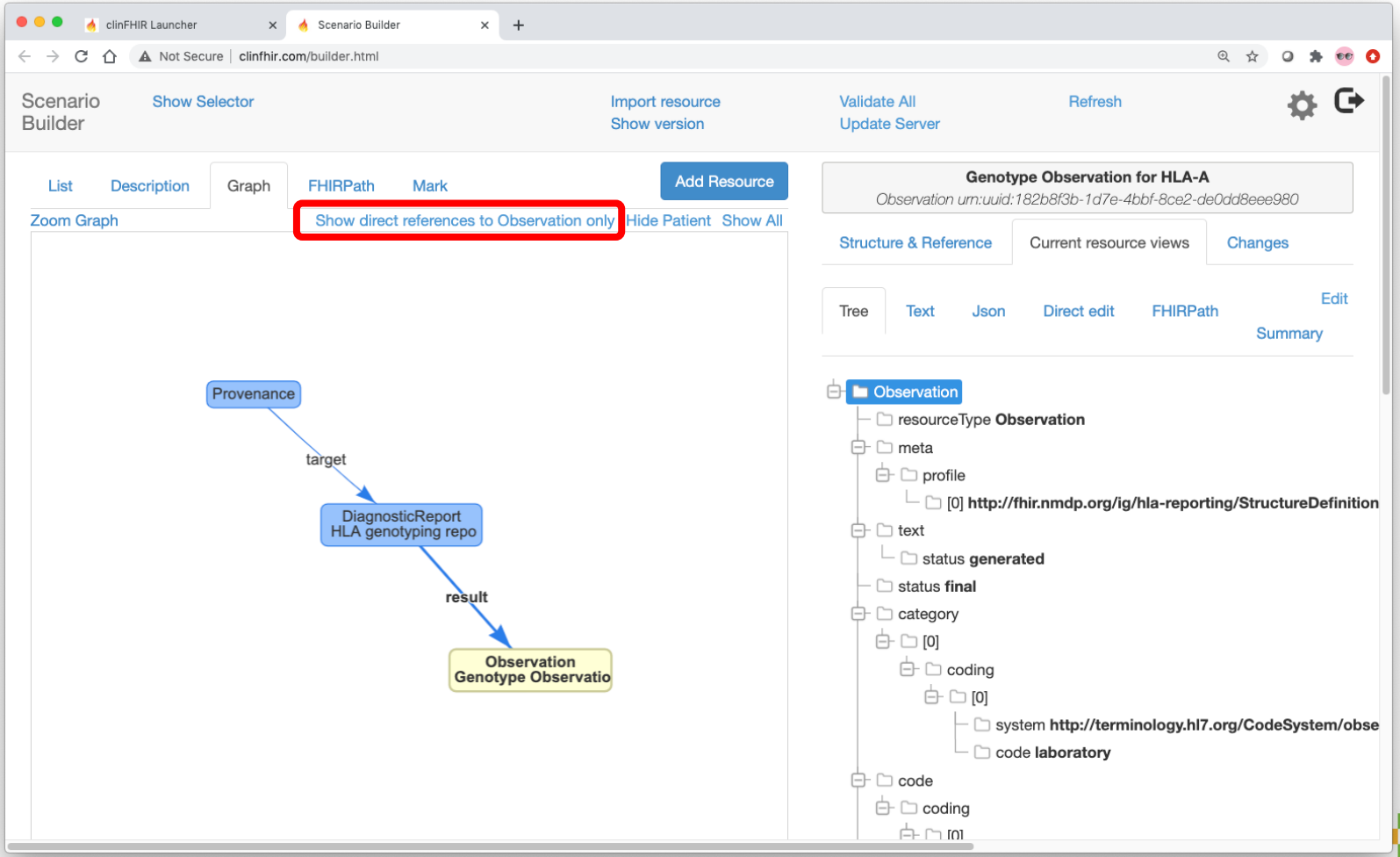

clinFHIR Launcher
Scenario Builder
Not Secure | clinfhir.com/builder.html
Scenario Builder
Hide Selector
Import resource
Validate All
Refresh
Local Scenarios
New Scenario
List
HLA-A
HLA-A from TARR
Type
Import Resource/s from text
Cancel
Import
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"meta": {"profile": ["http://fhir.nmdp.org/ig/hla-reporting/StructureDefinition/hla-molecularsequence"]},
"text": {
"status": "generated",
"div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"> HLA-A*24:02:01:01<\/div>"
},
Import as Logical Model
Strip text
Paste the XML or Json into the box and click the Import button (top right).
Json can be a single resource, a bundle of resources or a Logical Model. A Bundle must have 'fullUrl' set on each resource
For Logical Models, make sure to check the 'Logical Model' checkbox, or it will be imported as a StructureDefinition
XML must be a bundle, with a type of 'collection', no Bundle Id, and the conformance server must support the Bundle endpoint for both Json and XML

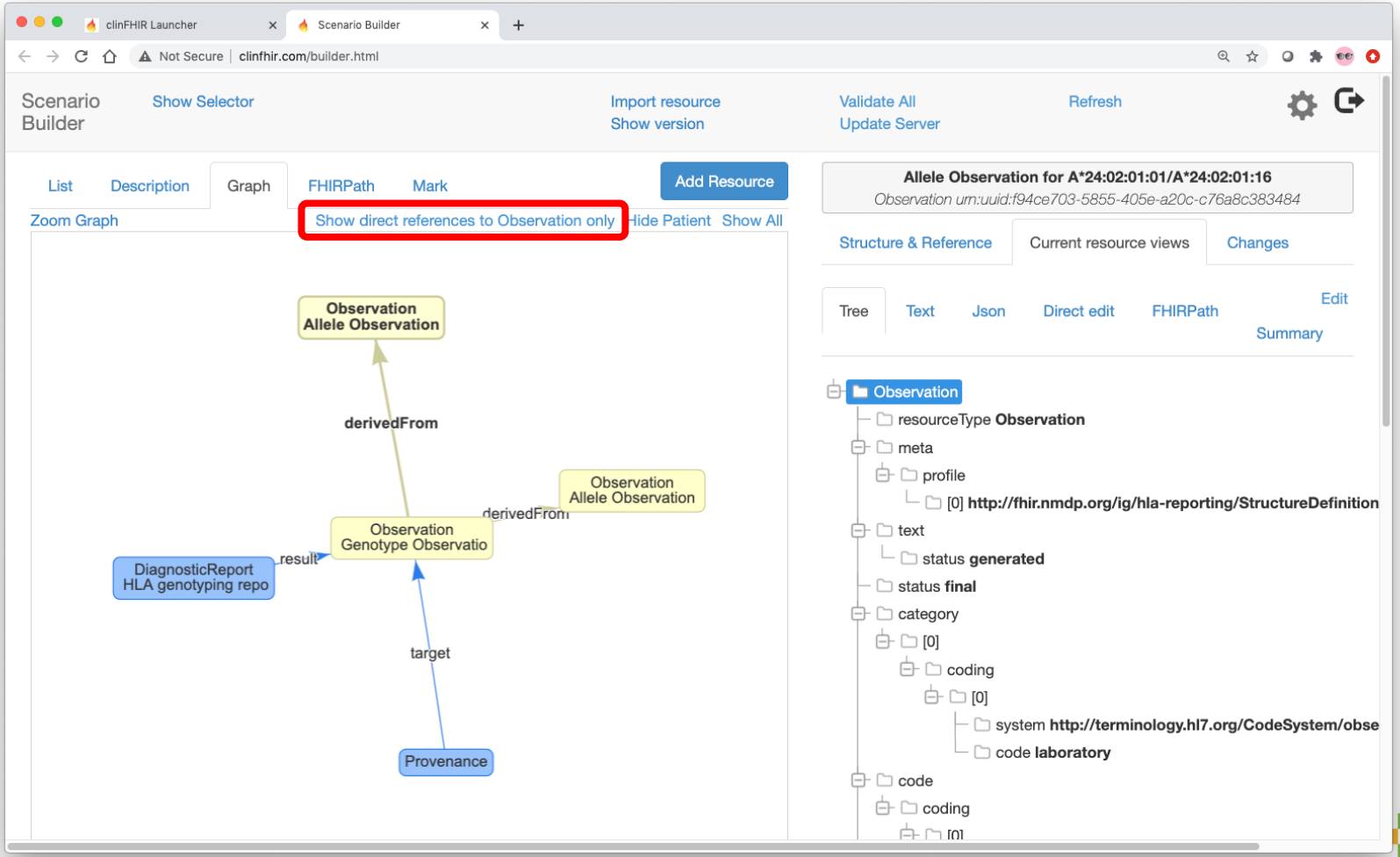

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Scenario Builder
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Scenario Builder
Hide Selector
Import resource
Validate All
Refresh
{"msg":"No matching profile with the canonical url: http://fhir.nmdp.org/ig/hla-reporting/StructureDefinition/hla-molecularsequence found on http://hapi.fhir.org/baseR4/StructureDefinition (but not a server error - just not present)"}
Ok
Local Scenarios
New Scenario
HLA-A
HLA-A from TARR
List
HLA-A*24:02:01:01
Current resource views
Changes
Toggle Input Mode
HLA-A*24:02:01:01
Type
Text
Device
software device: Tarr2Fhir
DiagnosticReport
HLA genotyping report for sample name = 03-
MolecularSequence
HLA-A*02:01:01:01
MolecularSequence
HLA-A*24:02:01:01
Observation
Genotype Observation for HLA-A
Observation
Allele Observation for A*24:02:01:01/A*24:02:01:16
Observation
Allele Observation for A*02:01:01:01/A*02:01:01:16/A*02:01:01:31/A*02:01:01:5
Provenance

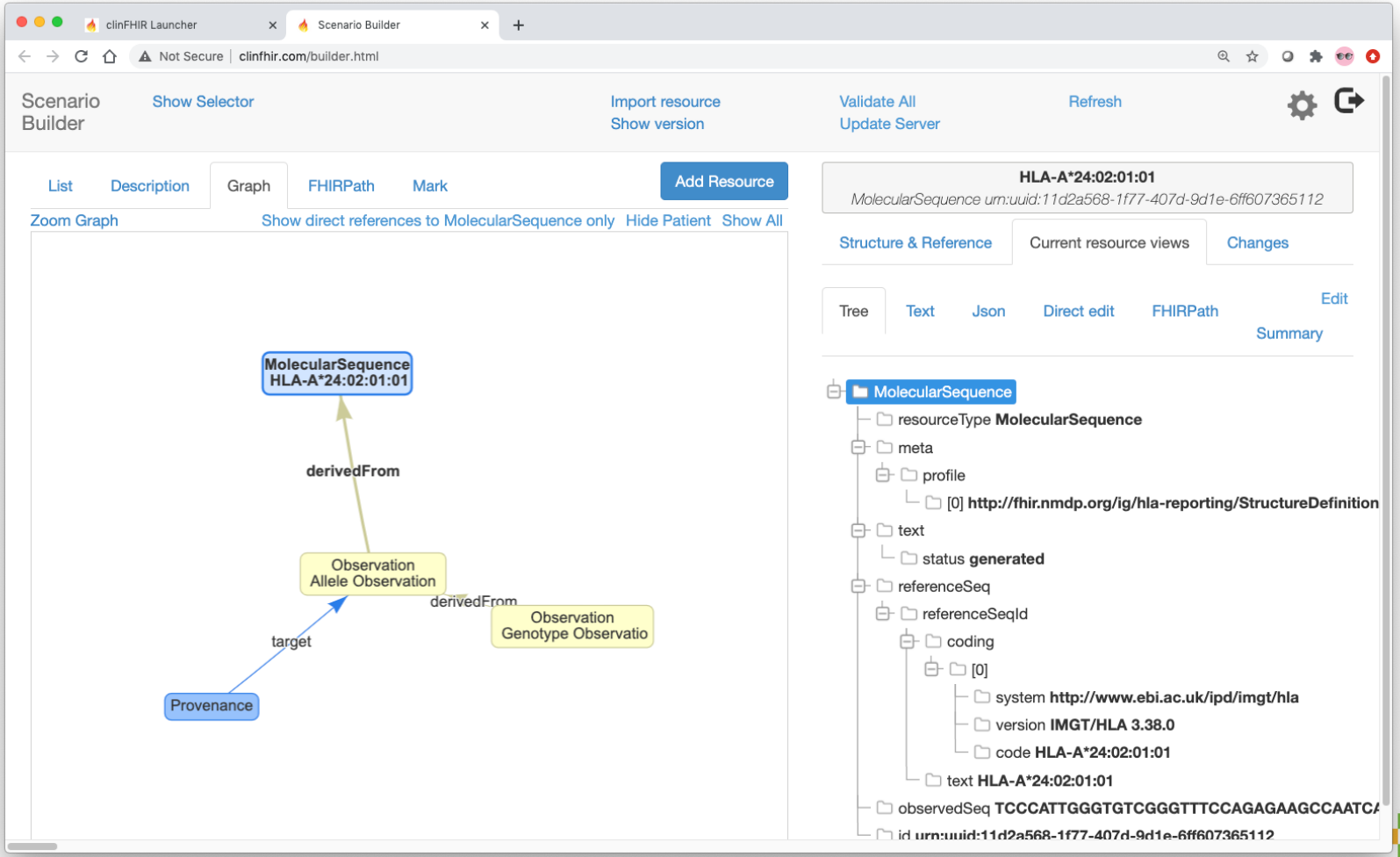

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Hide Selector
Import resource
Show version
Validate All
Update Server
Refresh
Local Scenarios
New Scenario
HLA-A
HLA-A from TARR
List
Description
Graph
FHIRPath
Add Resource
Mark
HLA-A*24:02:01:01
MolecularSequence urn:uuid:11d2a568-1f77-407d-9d1e-6ff607365112
Structure & Reference
Current resource views
Changes
Toggle Input Mode
HLA-A*24:02:01:01
List
Report
Type
Text
Device
software device: Tarr2Fhir
DiagnosticReport
HLA genotyping report for sample name = 03-
MolecularSequence
HLA-A*02:01:01:01
MolecularSequence
HLA-A*24:02:01:01
Observation
Genotype Observation for HLA-A
Observation
Allele Observation for A*24:02:01:01/A*24:02:01:16
Observation
Allele Observation for A*02:01:01:01/A*02:01:01:16/A*02:01:01:31/A*02:01:01:
Provenance


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Hide Selector
Import resource
Show version
Validate All
Update Server
Refresh
Local Scenarios
New Scenario
HLA-A
HLA-A from TARR
List
Description
Graph
FHIRPath
Add Resource
Mark
Zoom Graph
Hide Patient
Show direct references to MolecularSequence only
HLA-A*24:02:01:01
MolecularSequence urn:uuid:11d2a568-1f77-407d-9d1e-6ff607365112
Structure & Reference
Current resource views
Changes
Toggle Input Mode
HLA-A*24:02:01:01
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Scenario Builder
Show Selector
Import resource
Show version
Validate All
Update Server
Refresh
List
Description
Graph
FHIRPath
Mark
Add Resource
Zoom Graph
Show direct references to DiagnosticReport only
Device software device: Ta
MolecularSequence HLA-A*24:02:01:01
MolecularSequence HLA-A*02:01:01:01
Observation Allele Observation
Observation Genotype Observatio
Observation Allele Observation
DiagnosticReport HLA genotyping repo
Provenance
target
derivedFrom
HLA genotyping report for sample name = 03-
DiagnosticReport urn:uuid:1eec6a88-ae77-4a72-9629-580d269e21b3
Structure & Reference
Current resource views
Changes
Tree
Text
Json
Direct edit
FHIRPath
Edit
Summary
DiagnosticReport
resourceType DiagnosticReport
meta
profile
[0] http://fhir.nmdp.org/ig/hla-reporting/StructureDefinition
text
status generated
extension
[0]
url http://fhir.nmdp.org/ig/hla-reporting/StructureDefinition
valueCodeableConcept
coding
[0]
system http://glstring.org
code hla#3.38.0#HLA-A*02:01:01:01/HLA-A*02:01
status final
code


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FHIRPath
Mark
Add Resource
Zoom Graph
Show direct references to Observation only
Hide Patient
Show All
Provenance
target
DiagnosticReport
HLA genotyping repo
result
Observation
Genotype Observatio
Genotype Observation for HLA-A
Observation urn:uuid:182b8f3b-1d7e-4bbf-8ce2-de0dd8eee980
Structure & Reference
Current resource views
Changes
Tree
Text
Json
Direct edit
FHIRPath
Edit
Summary
Observation
resourceType Observation
meta
profile
[0] http://fhir.nmdp.org/ig/hla-reporting/StructureDefinition
text
status generated
status final
category
[0]
coding
[0]
system http://terminology.hl7.org/CodeSystem/obse
code laboratory
code
coding
[0]


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Mark
Add Resource
Zoom Graph
Show direct references to Observation only
Hide Patient
Show All
Observation
Allele Observation
derivedFrom
Observation
Allele Observation
derivedFrom
Observation
Genotype Observatio
result
DiagnosticReport
HLA genotyping repo
target
Provenance
Allele Observation for A*24:02:01:01/A*24:02:01:16
Observation urn:uuid:f94ce703-5855-405e-a20c-c76a8c383484
Structure & Reference
Current resource views
Changes
Tree
Text
Json
Direct edit
FHIRPath
Edit
Summary
Observation
resourceType Observation
meta
profile
[0] http://fhir.nmdp.org/ig/hla-reporting/StructureDefinition
text
status generated
status final
category
[0]
coding
[0]
system http://terminology.hl7.org/CodeSystem/obse
code laboratory
code
coding
[0]


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Scenario Builder
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Scenario Builder
Show Selector
Import resource
Show version
Validate All
Update Server
Refresh
List
Description
Graph
FHIRPath
Mark
Add Resource
Zoom Graph
Show direct references to MolecularSequence only
Hide Patient
Show All
MolecularSequence HLA-A*24:02:01:01
derivedFrom
Observation Allele Observation
derivedFrom
Observation Genotype Observatio
target
Provenance
HLA-A*24:02:01:01
MolecularSequence urn:uuid:11d2a568-1f77-407d-9d1e-6ff607365112
Structure & Reference
Current resource views
Changes
Tree
Text
Json
Direct edit
FHIRPath
Edit
Summary
MolecularSequence
resourceType MolecularSequence
meta
profile
[0] http://fhir.nmdp.org/ig/hla-reporting/StructureDefinition
text
status generated
referenceSeq
referenceSeqId
coding
[0]
system http://www.ebi.ac.uk/ipd/imgt/hla
version IMGT/HLA 3.38.0
code HLA-A*24:02:01:01
text HLA-A*24:02:01:01
observedSeq TCCCATTGGGTGTCGGGTTTCCAGAGAAGCCAATCA
id urn:uuid:11d2a568-1f77-407d-9d1e-6ff607365112

TARR2FHIR

- Provides an API through a REST service which can be invoked using tools such as Postman, cURL, or a programming/scripting language.

- Does not address reporting of novel variants. This is because of these variants are reported in the TARR file using coordinates based on a proprietary multi-sequence alignment of a set of HLA alleles.
- Need to configure Direct FHIR API to receive HLA Bundles (these can be fairly complex).

Thank you!